

DISTRIBUTIONAL GROWTH ACCOUNTING: EDUCATION AND THE REDUCTION OF GLOBAL POVERTY, 1980-2019

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SUPPLEMENTARY ONLINE APPENDIX

Abstract

This appendix supplements my article “Distributional Growth Accounting: Education and the Reduction of Global Poverty, 1980-2019.” It contains additional methodological details, as well as supplementary figures and tables.

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A. Additional Key Figures and Tables

Table A1 – Actual, Counterfactual, and True Returns to Schooling by World Region

	Primary		Secondary		Tertiary		Implied Return Per Year of Schooling		
	2019 Return	Counterfactual Return	2019 Return	Counterfactual Return	2019 Return	Counterfactual Return	2019 Return	True Return	Counterfactual Return
World Average	25%	48%	42%	89%	86%	134%	7.8%	11.0%	13.9%
Europe / Northern America	41%	51%	54%	127%	66%	122%	9.3%	13.0%	15.5%
Latin America	45%	61%	45%	108%	103%	114%	7.1%	9.4%	11.3%
China	12%	39%	49%	117%	95%	143%	8.3%	12.8%	15.9%
India	17%	32%	27%	53%	103%	166%	8.0%	10.8%	14.0%
Other Asia-Pacific	19%	47%	37%	67%	66%	123%	6.1%	8.7%	11.4%
Middle East and North Africa	19%	48%	24%	61%	42%	71%	4.4%	6.9%	10.2%
Sub-Saharan Africa	50%	81%	54%	96%	109%	149%	9.6%	12.3%	16.0%

Notes. The table reports actual, counterfactual, and true returns to schooling by world region and education level. Columns 2 to 7 report returns observed in 2019, $r_j(A^{2019}, L^{2019})$, versus counterfactual returns that would prevail absent educational expansion, $r_j(A^{2019}, L^{1980})$. The last three columns report the implied returns per year of schooling, calculated as $r_{\text{annual}}^*(A^{2019}) = 100 \times \frac{\ln H(A^{2019}, L^{2019}) - \ln H(A^{2019}, L^{1980})}{S^{2019} - S^{1980}}$ with S average years of schooling of the working-age population as reported in the Barro-Lee database. In other words, the 2019 implied return is the return per year that can reproduce the decline in output observed when using 2019 returns by level to construct the counterfactual. The implied true return per year of schooling, $r_{\text{annual}}^*(A^{2019})$, is the return that can reproduce the decline in output predicted by the model. All figures are population-weighted averages of returns estimated in each country.

Table A2 – Distributional Growth Accounting, World, 1980-2019: World Bank Data

	Annual Income Growth (%)	Contribution of Education (pp.)	Share of Growth Explained (%)
Full Population	1.4%	0.9	67%
Bottom 50%	2.7%	1.2	43%
Bottom 20%	2.5%	1.2	47%
Next 30%	2.8%	1.2	43%
Middle 40%	1.4%	1.2	82%
Top 10%	1.2%	0.7	62%
Top 1%	1.6%	0.6	35%
Top 0.1%	2.1%	0.5	23%

Notes. The table reports actual real annual income growth rates, the contribution of education to growth, and the corresponding share of growth explained by education for different groups of the world distribution of income. The income concept is per-capita pretax income, estimated by combining disposable income and consumption distributions from the World Bank data portal with estimates of the distribution of direct taxes and government transfers from [Gethin \(2024\)](#) and [Fisher-Post and Gethin \(2023\)](#).

Table A3 – Education and Global Poverty Reduction, 1980-2019: WID Data

	1980	2019	Difference (%)	Share of Decline Explained (%)
Global Poverty: \$2.15 / Day				
Actual	21%	9%	-58%	
Counterfactual	21%	14%	-36%	39%
Global Poverty: \$3.65 / Day				
Actual	42%	15%	-65%	
Counterfactual	42%	23%	-46%	30%
Global Poverty: \$6.85 / Day				
Actual	58%	29%	-50%	
Counterfactual	58%	42%	-28%	44%

Notes. The table compares the actual evolution of the global poverty headcount ratio to the evolution it would have followed absent educational expansion since 1980. All global poverty headcount ratios calculated using 2017 PPP USD. The income concept is pretax national income, as reported in the World Inequality Database. See Table IV for comparable results using per-capita consumption distributions from the World Bank.

Table A4 – From Standard to Distributional Growth Accounting
(Not All of Mixed Income Affected by Education)

		Share of Growth Explained, 1980-2019	
		Global Average	Global Bottom 20%
	Standard Growth Accounting		
(1)	Cross-Country Data, 10% Return	33%	16%
(2)	+ Adjusted Income Concept	41%	22%
(3)	+ Within-Country Inequality	41%	33%
(4)	+ Heterogeneous Labor Shares	41%	45%
(5)	+ Microdata	30%	25%
(6)	+ Supply Effects: Distributional Component	30%	38%
(7)	+ Supply Effects: Aggregate Component	42%	46%

Notes. This table reproduces Table V but assumes that only 75% of mixed income is affected by education in rows 2 to 7. This amounts to using the benchmark labor shares estimated in [Bachas et al. \(2022\)](#), which attribute 75% of mixed income to labor income, instead of labor shares including all of mixed income as in Table V.

Table A5 – Education and Inequality Between and Within Countries, 1980-2019

	1980	2019	Difference
Theil Index of Global Inequality			
Actual	1.09	1.09	0.00
Counterfactual	1.09	1.32	0.22
Between-Country Component			
Actual	0.63	0.36	-0.27
Counterfactual	0.63	0.37	-0.26
Within-Country Component			
Actual	0.46	0.73	0.27
Counterfactual	0.46	0.95	0.48
Within-Country Share (%)			
Actual	42%	67%	25
Counterfactual	42%	72%	30

Notes. The table compares the actual evolution of global inequality since 1980 to the evolution it would have followed absent educational expansion, decomposing these transformations into a between-country component and a within-country component. Within-country share: share of global inequality explained by inequality within countries.

Table A6 – Distributional Growth Accounting by World Region and Country Income Group, 1980-2019

	Full Population			Bottom 50%		
	Annual Income Growth (%)	Contribution of Education (pp.)	Share of Growth Explained (%)	Annual Income Growth (%)	Contribution of Education (pp.)	Share of Growth Explained (%)
Average Country	2.9%	0.7	24%	2.4%	1.2	49%
Europe / Northern America	1.5%	0.8	50%	0.9%	1.4	>100%
Latin America	0.6%	0.8	>100%	0.9%	1.4	>100%
China	6.1%	0.8	14%	4.6%	1.5	32%
India	4.0%	0.6	16%	2.8%	0.9	34%
Other Asia-Pacific	2.8%	0.7	25%	2.7%	1.1	39%
Middle East and North Africa	1.5%	0.5	31%	1.7%	0.8	45%
Sub-Saharan Africa	1.0%	0.8	74%	1.3%	1.0	77%
Low-income	0.9%	0.8	89%	1.3%	1.0	78%
Low-middle-income	3.0%	0.6	19%	2.5%	0.9	35%
High-middle-income	3.9%	0.8	21%	3.2%	1.4	44%
High-income	1.7%	0.7	45%	1.0%	1.4	>100%

Notes. The table reports actual real income growth rates, the contribution of education, and the corresponding share of growth that can be explained by education, for the full population and the poorest 50%. Population-weighted averages of country-specific growth rates and contributions of education by world region, by country income group, and in the world as a whole (average country).

Table A7 – Distributional Growth Accounting, World, 2000-2019

	Annual Income Growth (%)	Contribution of Education (pp.)	Share of Growth Explained (%)
Full Population	2.1%	0.5	22%
Bottom 50%	3.1%	1.0	33%
Bottom 20%	2.8%	1.1	42%
Next 30%	3.2%	1.0	31%
Middle 40%	2.9%	0.7	23%
Top 10%	1.5%	0.3	18%
Top 1%	1.9%	0.09	5.0%
Top 0.1%	2.2%	0.01	0.4%

Notes. The table reports actual real annual income growth rates, the contribution of education to growth, and the corresponding share of growth explained by education for different groups of the world distribution of income over the 2000-2019 period.

Table A8 – Distributional Growth Accounting by Skill Group, 1980-2019:
Actual Minus Counterfactual Income by Skill Group, Average Country (%)

	Actual Minus Counterfactual Income (%)		
	Share of Workers (%)	Without Imperfect Substitution	With Imperfect Substitution
Always No Schooling	12.1%	0%	-22.1%
Newly No Schooling	21.3%	-34.5%	-49.2%
Always Primary	6.4%	0%	-16.5%
Newly Primary	27.9%	-34.9%	-45.3%
Always Secondary	11.5%	0%	6.5%
Newly Secondary	12.8%	-39.5%	-33.4%
Always Tertiary	7.4%	0%	30.7%
Average Income		-23.9%	-29.8%

Notes. The table reports how much lower or higher the wages of different skill groups would be in the average country (population-weighted) if education had not improved: $\frac{w_j(A^{2019}, L^{1980}) - w_j(A^{2019}, L^{2019})}{w_j(A^{2019}, L^{2019})}$. Gains or losses from educational expansion are reported separately for the newly unskilled (who see their education decline in the counterfactual) and the always skilled or unskilled (who do not see their education change in the counterfactual). Without imperfect substitution: results without supply effects. With imperfect substitution: incorporate supply effects.

Table A9 – Returns to Schooling by World Region, 2000-2019

	Primary		Secondary		Tertiary	
	2000	2019	2000	2019	2000	2019
Average Country	59%	20%	89%	40%	94%	84%
Europe / Northern America			59%	53%	61%	64%
Latin America	67%	43%	78%	44%	115%	98%
China	67%	12%	146%	49%	174%	95%
India	56%	17%	119%	27%	85%	103%
Other Asia-Pacific	50%	19%	54%	38%	62%	69%
Middle East and North Africa	35%	19%	26%	18%	48%	38%
Sub-Saharan Africa	72%	35%	68%	54%	95%	95%

Notes. The table reports estimates of the returns to schooling $r_j(A^t, L^t)$ by world region and in the average country in 2000 and 2019, re-expressed as percentage differences to ease interpretation. Population-weighted averages of returns to schooling estimated in each country. The table covers 109 countries with data on incomes and education around both 2000 and 2019.

B. Additional Methodological Details

This section presents the steps followed to solve the model and estimate the total contribution of education to growth by income group in each country.

B.1. Solving the Model

I start by solving the model. The production function is:

$$L = \left(A_{\overline{ter}} L_{\overline{ter}}^{\frac{\sigma_1-1}{\sigma_1}} + A_{ter} L_{ter}^{\frac{\sigma_1-1}{\sigma_1}} \right)^{\frac{\sigma_1}{\sigma_1-1}} \quad (B1)$$

$$L_{\overline{ter}} = \left(A_{\overline{sec}} L_{\overline{sec}}^{\frac{\sigma_2-1}{\sigma_2}} + A_{sec} L_{sec}^{\frac{\sigma_2-1}{\sigma_2}} \right)^{\frac{\sigma_2}{\sigma_2-1}} \quad (B2)$$

$$L_{\overline{sec}} = \left(A_{non} L_{non}^{\frac{\sigma_3-1}{\sigma_3}} + A_{pri} L_{pri}^{\frac{\sigma_3-1}{\sigma_3}} \right)^{\frac{\sigma_3}{\sigma_3-1}} \quad (B3)$$

Where country and time subscripts are omitted to simplify the exposition. At the upper level, output is produced by combining tertiary-educated and non-tertiary-educated workers L_{ter} and $L_{\overline{ter}}$. The intermediate level includes workers with secondary L_{sec} and below-secondary education $L_{\overline{sec}}$. Finally, the lower level includes workers with primary education L_{pri} and no schooling L_{non} .

Wages Profit maximization implies the following wages:

$$W_{\overline{ter}} = L^{\frac{1}{\sigma_1}} A_{\overline{ter}} L_{\overline{ter}}^{-\frac{1}{\sigma_1}} \quad (B4)$$

$$W_{ter} = L^{\frac{1}{\sigma_1}} A_{ter} L_{ter}^{-\frac{1}{\sigma_1}} \quad (B5)$$

$$W_{\overline{sec}} = L^{\frac{1}{\sigma_1}} A_{\overline{ter}} L_{\overline{ter}}^{-\frac{1}{\sigma_1}} L_{\overline{ter}}^{\frac{1}{\sigma_2}} A_{\overline{sec}} L_{\overline{sec}}^{-\frac{1}{\sigma_2}} \quad (B6)$$

$$W_{sec} = L^{\frac{1}{\sigma_1}} A_{\overline{ter}} L_{\overline{ter}}^{-\frac{1}{\sigma_1}} L_{\overline{ter}}^{\frac{1}{\sigma_2}} A_{sec} L_{sec}^{-\frac{1}{\sigma_2}} \quad (B7)$$

$$W_{non} = L^{\frac{1}{\sigma_1}} A_{\overline{ter}} L_{\overline{ter}}^{-\frac{1}{\sigma_1}} L_{\overline{ter}}^{\frac{1}{\sigma_2}} A_{\overline{sec}} L_{\overline{sec}}^{-\frac{1}{\sigma_2}} L_{\overline{sec}}^{\frac{1}{\sigma_3}} A_{non} L_{non}^{-\frac{1}{\sigma_3}} \quad (B8)$$

$$W_{pri} = L^{\frac{1}{\sigma_1}} A_{\overline{ter}} L_{\overline{ter}}^{-\frac{1}{\sigma_1}} L_{\overline{ter}}^{\frac{1}{\sigma_2}} A_{\overline{sec}} L_{\overline{sec}}^{-\frac{1}{\sigma_2}} L_{\overline{sec}}^{\frac{1}{\sigma_3}} A_{pri} L_{pri}^{-\frac{1}{\sigma_3}} \quad (B9)$$

Returns to Schooling The returns to schooling are:

$$r_{pri} = \ln\left(\frac{w_{pri}}{w_{non}}\right) = \ln\left(\frac{A_{pri}}{A_{non}}\right) - \frac{1}{\sigma_3} \ln\left(\frac{L_{pri}}{L_{non}}\right) \quad (B10)$$

$$r_{sec} = \ln\left(\frac{w_{sec}}{w_{pri}}\right) = \ln\left(\frac{A_{sec}}{A_{sec}A_{pri}}\right) - \frac{1}{\sigma_2} \ln\left(\frac{L_{sec}}{L_{sec}}\right) - \frac{1}{\sigma_3} \ln\left(\frac{L_{sec}}{L_{pri}}\right) \quad (B11)$$

$$r_{ter} = \ln\left(\frac{w_{ter}}{w_{sec}}\right) = \ln\left(\frac{A_{ter}}{A_{ter}A_{sec}}\right) - \frac{1}{\sigma_1} \ln\left(\frac{L_{ter}}{L_{ter}}\right) - \frac{1}{\sigma_2} \ln\left(\frac{L_{ter}}{L_{sec}}\right) \quad (B12)$$

B.2. Estimation

The Barro-Lee database allows me to observe L_{non} , L_{pri} , L_{sec} , and L_{ter} .¹ Meanwhile, the microdata allow me to estimate the returns to schooling r_{pri} , r_{sec} , and r_{ter} . The only elements that remain to be estimated are L_{sec} , L_{ter} , and the technology parameters A_{non} , A_{pri} , A_{sec} , A_{sec} , A_{ter} , and A_{ter} .

Let us assume without loss of generality that $A_{ter} = 1 - A_{ter}$, $A_{sec} = 1 - A_{sec}$, and $A_{non} = 1 - A_{pri}$. All parameters can then be recovered as follows.

First, rearranging equation B10, I recover A_{non} and A_{pri} from the return to primary education:

$$A_{non} = \frac{1}{1 + \exp\left(r_{pri} + \frac{1}{\sigma_3} \ln\left(\frac{L_{pri}}{L_{non}}\right)\right)} \quad (B13)$$

$$A_{pri} = 1 - A_{non} \quad (B14)$$

Second, I construct L_{sec} :

$$L_{sec} = \left(A_{non} L_{non}^{\frac{\sigma_3-1}{\sigma_3}} + A_{pri} L_{pri}^{\frac{\sigma_3-1}{\sigma_3}}\right)^{\frac{\sigma_3}{\sigma_3-1}} \quad (B15)$$

Third, I recover A_{sec} and A_{sec} from the return to secondary education:

$$A_{sec} = \frac{1}{1 + A_{pri} \exp\left(r_{sec} + \frac{1}{\sigma_2} \ln\left(\frac{L_{sec}}{L_{sec}}\right) + \frac{1}{\sigma_3} \ln\left(\frac{L_{sec}}{L_{pri}}\right)\right)} \quad (B16)$$

$$A_{sec} = 1 - A_{sec} \quad (B17)$$

¹In a handful of cases, the share of workers with primary education or no schooling dropped to zero, leading counterfactual returns to schooling to diverge to infinity. To avoid these extreme cases, I impose a lower bound on the share of workers with primary education and no schooling of 1%.

Fourth, I construct $L_{\overline{ter}}$:

$$L_{\overline{ter}} = \left(A_{\overline{sec}} L_{\overline{sec}}^{\frac{\sigma_2-1}{\sigma_2}} + A_{sec} L_{sec}^{\frac{\sigma_2-1}{\sigma_2}} \right)^{\frac{\sigma_2}{\sigma_2-1}} \quad (B18)$$

Fifth, I recover $A_{\overline{ter}}$ and A_{ter} from the return to tertiary education:

$$A_{\overline{ter}} = \frac{1}{1 + A_{sec} \exp\left(r_{ter} + \frac{1}{\sigma_1} \ln\left(\frac{L_{ter}}{L_{\overline{ter}}}\right) + \frac{1}{\sigma_2} \ln\left(\frac{L_{\overline{ter}}}{L_{sec}}\right)\right)} \quad (B19)$$

$$A_{ter} = 1 - A_{\overline{ter}} \quad (B20)$$

Sixth, I construct L :

$$L = \left(A_{\overline{ter}} L_{\overline{ter}}^{\frac{\sigma_1-1}{\sigma_1}} + A_{ter} L_{ter}^{\frac{\sigma_1-1}{\sigma_1}} \right)^{\frac{\sigma_1}{\sigma_1-1}} \quad (B21)$$

All the parameters are now known: counterfactual output and wages can be directly calculated by plugging the 1980 distribution of educational attainment into equations B1 to B3.

B.3. Bringing the Model to the Microdata

Once all the parameters of the model are estimated, I bring it to the microdata. I construct the distributional growth accounting decomposition in four steps.

1) Downgrade Education Levels The starting point is individual-level data on wages and education. In each survey, I keep individuals aged 25 to 65, with no missing information on education and with positive personal income. Next, I match the microdata with information on the distribution of educational attainment of the working-age population by age group and gender in 1980 from the Barro-Lee database, covering four education levels: no schooling, incomplete or complete primary education, incomplete or complete secondary education, and incomplete or complete tertiary education. To move from observed educational attainment to counterfactual educational attainment, I randomly sample individuals and downgrade their education levels until matching 1980 totals in each country.

Individuals belonging to closest education categories are given priority in the simulation. For instance, if 20% of individuals had no schooling in 1980, compared to 10% today, I randomly sample 10% of individuals among the primary education group and downgrade their education level to no schooling. When closest

education levels do not contain enough individuals (e.g., only 5% of individuals in this survey have primary education), I instead sample individuals from the category above (secondary education in this example). The outcome is a modified survey, in which the distribution of education corresponds to that observed in 1980. This survey contains “non-treated” observations, corresponding to individuals with unchanged education, as well as “treated” individuals whose education has been downgraded. This approach is very similar to the one recently adopted by [Hershbein, Kearney, and Pardue \(2020\)](#) to estimate the distributional effects of expanding access to college in the United States.

2) Adjust Wages The second step is to adjust the wages of “treated” and “non-treated” individuals. Since we are moving from the 2019 to the 1980 distribution of wages, the relevant parameter is the gap between the 2019 initial wage and the 1980 target wage. More specifically, consider an individual i with initial personal income $y_i(A^{2019}, L^{2019})$, whose education level is downgraded from s_2 to s_1 . Their counterfactual income absent educational expansion is:

$$y_i(A^{2019}, L^{1980}) = \exp \left[\ln \left(y_i(A^{2019}, L^{2019}) \right) - \ln \left(\frac{w_{s_2}(A^{2019}, L^{2019})}{w_{s_1}(A^{2019}, L^{1980})} \right) \right] \quad (\text{B22})$$

Put simply, individual i see their income move from the actual wage of skill group s_2 to the counterfactual wage of skill group s_1 . Actual and counterfactual wages are estimated above, so the calculation of this return is straightforward. For individuals whose education is downgraded by several levels, I use the corresponding cumulative returns. For instance, an individual downgraded from secondary education to no schooling will see their income decline by a return equal to $\frac{w_{sec}(A^{2019}, L^{2019})}{w_{non}(A^{2019}, L^{1980})}$.

Notice that because of imperfect substitution, non-treated workers are also affected by educational expansion. Typically, because educational expansion is associated with an increase in the supply of skilled workers, this implies that non-treated higher-educated workers see their wage increase, while non-treated lower-educated workers see their wage decrease in the counterfactual.

3) Derivation of Total Income The next step is to move from this counterfactual distribution of labor income to a counterfactual distribution of total income. I first aggregate actual labor income $y_L(A^{2019}, L^{2019})$ and counterfactual labor income $y_L(A^{2019}, L^{1980})$ in the survey microdata by decile, and calculate the

corresponding log difference between actual income and counterfactual income by decile:

$$\psi^d = \ln y_L^d(A^{2019}, L^{2019}) - \ln y_L^d(A^{2019}, L^{1980}) \quad (B23)$$

This yields a measure of how much lower labor income would be if education had not improved.

I then incorporate these estimates into global income distribution data. I start with distributions from the World Inequality Database, which provide information on the average pretax income of each generalized percentile (all percentiles from p0 to p99, followed by a further decomposition of top incomes up to p99.999p100). I merge estimates of ψ^d by country-year-decile with this database.² I then calculate counterfactual total pretax income of generalized percentile p as:

$$y^p(A^{2019}, L^{1980}) = \exp\left(\ln y^p(A^{2019}, L^{2019}) - \alpha^p \psi^p\right) \quad (B24)$$

Where α^p is the capital income share of percentile p . Finally, I construct separate actual and counterfactual world distributions of income from 1980 to 2019, by ranking all individuals in the world by each income concept and aggregating average income by global generalized percentile.

4) Growth Accounting The final step is to calculate the share of growth accounted for by education for income group p :

$$\text{Share}_{L^{\text{p,total}}} = \frac{\ln y^p(A^{2019}, L^{2019}) - \ln y^p(A^{2019}, L^{1980})}{\ln y^p(A^{2019}, L^{2019}) - \ln y^p(A^{1980}, L^{1980})} \quad (B25)$$

²To get smoother profiles of counterfactual income by generalized percentile, I assume that ψ^d for each decile corresponds to the ratios observed for p5, p15, p25, p35, p45, p55, p65, p75, p85, and p95. I make the simplifying assumption that deciles of labor income are the same as deciles of total pretax income. I then interpolate ψ^d between percentiles to fill in missing values. I assume that values observed for percentiles within the bottom 5% and the top 5% are those observed for p5 and p95, respectively. Finally, I drop the bottom 5% in each country, which is by convention coded as having zero income in the World Inequality Database.

C. Natural Experiments

This appendix exploits evidence from three natural experiments to shed light on the aggregate and distributional effects of schooling. Section C.1 outlines the general econometric framework. Sections C.2, C.3, and C.4 turn to analyzing the Indian District Primary Education Program, the Indonesian INPRES school construction program, and U.S. compulsory schooling laws.

C.1. General Methodology

A large literature focuses on causally identifying individual returns to schooling. Less is known of its distributional effects at the level of regions or countries. This section attempts to shed some light on these effects in the context of three natural experiments. Consider the following empirical specification:

$$\ln y_{rt}^i = \gamma_0^i + \gamma_1^i S_{rt} + X_{rt} \beta + \delta_r + \delta_t + \varepsilon_{rt} \quad (\text{C1})$$

$$S_{rt} = \alpha_0 + \alpha_1 Z_{rt} + \eta_{rt} \quad (\text{C2})$$

Where i denotes income groups in subnational regions r at time t . The objective is to estimate the impact of increasing average regional schooling S_{rt} on $\ln y_{rt}^i$, the log average income of income group i . X_{rt} is a vector of controls, δ_r are subnational region fixed effects, and δ_t are time fixed effects.

The parameter of interest is γ_1^i , the semi-elasticity of average income of group i to regional average years of schooling. One option is to directly estimate equation C1 by OLS. Alternatively, average schooling S_{rt} can be instrumented using an instrument Z_{rt} , such as compulsory schooling laws. This approach has also been used to estimate human capital externalities, in particular in the case of U.S. compulsory schooling laws (Acemoglu and Angrist, 2000; Ciccone and Peri, 2006; Guo, Roys, and Seshadri, 2018). The main addition here is the focus on distributional effects, which amounts to estimating heterogeneous treatment effects by income group.

Estimating the distributional effects of educational expansion is empirically challenging, because it requires two sets of data that are rarely jointly available: data on the distribution of income within subnational regions, and an instrument that can predict quasi-random variation in regional schooling. Drawing on existing work, I study three such sources of variation: the India District Primary Education Program, the Indonesian School Construction Program, and U.S. state compulsory schooling laws.

C.2. India District Primary Education Program, 1994-2004

Context Between the 1990s and the beginning of the 2000s, India engaged in a massive expansion of public schooling, the District Primary Education Program (DPEP), targeting low-literacy regions. Districts with a female literacy rate below the national average were more likely to benefit from the policy. Exploiting this allocation rule, [Khanna \(2023\)](#) estimates the general equilibrium effects of the program using a regression discontinuity design. He finds a return to schooling of about 13% per year (after accounting for general equilibrium effects). General equilibrium effects induced by the greater relative supply of skilled workers depress returns by one-third, while indirectly benefiting unskilled workers.

Data and Empirical Specification I exploit data from the replication package provided by [Khanna \(2023\)](#). Exposure to the program is determined by district female literacy in 1991. Individual outcomes are obtained from the 2009 National Sample Survey (NSS), which covers wages and education at the district level. As in [Khanna \(2023\)](#), the sample is restricted to all adults aged 17 to 75 with positive wage income.

I then estimate the impact of the policy using the same regression discontinuity design as in the paper, comparing districts below and above the average female literacy rate. Optimal bandwidths are calculated using either the [Calonico, Cattaneo, and Titiunik \(2014\)](#) method or the [Imbens and Kalyanaraman \(2012\)](#) method (henceforth CCT and I and K, respectively). The main addition is that I focus on the effect of the program on the average wage of each wage quintile, yielding reduced-form estimates of the distributional incidence of primary education expansion.

Results Table [C1](#) presents the results. Increasing district average years of schooling by one year is associated with a 0.12 log-point increase in wages in treated districts (CCT method). This effect is almost two times larger for the bottom 20% of earners. In contrast, the top 20% see their average wage decline, although the coefficient is not statistically significant. Results relying on the I and K method are similar, but the aggregate effect of educational expansion appears even larger. Aggregate returns to schooling estimated using this method are in the range of individual returns estimated by [Khanna \(2023\)](#), who finds returns of 0.16 (CCT) to 0.21 (I and K) log points using conventional 2SLS estimates, and 0.13 (CCT) after accounting for general equilibrium effects.

Table [C2](#) compares the CCT estimates with simulated effects of expanding primary education, under different parametrizations of the return to schooling and the elasticity of substitution between skilled and

unskilled workers. The simulation is done by upgrading the education of randomly sampled individuals from no schooling to primary education, increasing their earnings using the return to schooling, and finally adjusting relative wages for general equilibrium effects.

Simulated estimates fall close to the true effects of the policy. With a return of 16%, increasing average education by one year is associated with an increase in average wages of about 12%, which is identical to the actual effects. Distributional are also very similar to those estimated with the RD design. Both in the simulation and in the natural experiment, benefits appear relatively similar for the first four quintiles and significantly lower for the top 20%. This can be rationalized by the fact that in India, workers with no schooling and workers with basic education are both prevalent among the bottom 80% of the distribution, so that upgrading some workers from no schooling to basic education benefits this entire group.

C.3. Indonesia School Construction Program, 1973-1978

Context Between 1973 and 1978, Indonesia engaged in a massive school construction program expanding access to basic education. Exploiting differences in exposure to newly built schools across cohorts and regions, [Duflo \(2001\)](#) estimates individual returns to schooling ranging from 7% to 11%. A number of studies have updated and extended her analysis since then, focusing on intergenerational effects ([Akresh, Halim, and Kleemans, 2023](#)), structural transformation ([Karachiwalla and Palloni, 2019](#)), or rural-urban migration ([Hsiao, 2023](#)). [Duflo \(2004\)](#) also moves beyond individual outcomes to focus on spillovers of the program to non-treated groups. Her analysis shows mixed findings, suggesting a decline in the wages of non-treated groups, but an increase in employment in the formal sector.

Data Drawing on the work of [Duflo \(2004\)](#), I exploit exposure to the program by district to estimate the aggregate and distributional effects of primary education expansion. My analysis expands her work in two ways. First, I expand the time coverage, which increases statistical power and allows me to get closer to long-run effects. To do so, I harmonize every round of the SUSENAS, a household survey covering about a million individuals every year, from 1993 to 2019. This allows me to construct a balanced panel of 230 districts with annual data on education, the distribution of consumption, and other sociodemographic variables.³ Second, I study the effects of the program on total district consumption and its distribution by consumption quintile, while [Duflo \(2004\)](#) focuses on spillover effects on older cohorts. The sample is

³Some districts have undergone splits and merges. I rely on crosswalks provided by [Roodman \(2022\)](#) to ensure consistent boundaries over time.

restricted to all adults aged 15 to 70. Consumption is split equally between all household members.

Empirical Specification The empirical specification is given by equation C1. I estimate the effect of average years of schooling in district r on the log average consumption of decile i , controlling for district and year fixed effects. Average years of schooling is instrumented by the interaction between survey years and the number of schools built per 5-14 population between 1974 and 1978, as in Duflo (2004).⁴ The school construction program is thus taken as an instrument for differential trends in the education of the working-age population across districts from 1993 to 2019. The identification assumption, analogous to Duflo (2004), is that there is no unobserved shock both correlated with the program and affecting household expenditure during that period.

Results Table C3 presents the main results. The baseline specification controls for the demographic and gender composition of each district, the share of college graduates, and district and year fixed effects. The aggregate return to schooling is about 9%. This effect is almost four times larger for the bottom quintile (0.22 log points) than for the top quintile (0.058). Columns 4 to 6 add controls for 1971 primary school enrollment and water and sanitation spending interacted with survey year, as in Duflo (2004). These estimates are underpowered, but the results are qualitatively similar. Columns 7 to 9 further add controls for 1971 child population and population density interacted with survey year. This model is even more underpowered, but the point estimates remain very similar. The coefficient on the average income of the bottom 20% remains large and statistically significant. While the sample size is not sufficient to precisely estimate aggregate returns to schooling, the progressive nature of the policy stands out across all specifications.

Table C4 compares the benchmark estimates to simulated effects of the policy using the 1996 Indonesian labor force survey (SAKERNAS). The simulation is done exactly as in the Indian case, upgrading the education of randomly sampled individuals from no schooling to primary education, increasing their earnings using the return to schooling, and finally adjusting relative wages. As in India, the simulation does a good job at reproducing results from the natural experiment. The expansion of primary education is estimated to be progressive in all specifications, with orders of magnitude similar to those found in the data.

⁴Given significant noise introduced by the low sample size available for each district-year cell, I specify survey years as a continuous variable in the first stage. Indeed, we should expect the program to have introduced smooth, secular differential trends in educational expansion. Constraining the interaction of survey years and treatment intensity to follow such a secular trend makes the results less sensitive to different empirical strategies.

C.4. U.S. Compulsory Schooling Laws, 1875-1961

Context Between the mid-19th and the mid-20th century, U.S. states gradually implemented laws limiting child labor and enforcing compulsory school attendance. The effects of these laws were first studied by [Acemoglu and Angrist \(2000\)](#), who combined data on 1914-1965 laws with census microdata to estimate the magnitude of human capital spillovers. Their analysis gave rise to a rich literature exploiting compulsory schooling laws to estimate individual returns to schooling ([Clay, Lingwall, and Stephens, 2021](#); [Stephens and Yang, 2014](#)), elasticities of substitution between skill groups ([Ciccone and Peri, 2006](#)), and human capital externalities ([Ciccone and Peri, 2006](#); [Guo, Roys, and Seshadri, 2018](#); [Iranzo and Peri, 2009](#)).

Data My analysis extends previous work in two ways. First, I study the aggregate and distributional effects of educational expansion, while existing studies focus on dimensions of these effects separately. Second, I exploit recently compiled data by [Clay, Lingwall, and Stephens \(2021\)](#), covering compulsory schooling laws over the entire 1875-1961 period. This represents an important improvement over the previous literature, which only covered laws implemented after 1915. I rely on the 1940 to 2000 census microdata samples available from IPUMS, which cover personal income, state of birth, state of residence, education, and other sociodemographic variables. The sample is restricted to all adults aged 25 to 65 with positive personal income (wage income in 1940) living in the contiguous United States.

Empirical Specification As in the Indian and Indonesian cases, I regress the average income of each personal income decile on average state schooling, instrumented by compulsory schooling laws. I use the following instrument for average years of schooling S_{st} in state s at time t :

$$S_{st} = \pi_0 + \pi_1 \sum_c \sum_{s'} N_{css't} RS_{cs'} + \theta_s + \theta_t + u_{st} \quad (C3)$$

$RS_{cs'}$ is required years of schooling for cohort c born in state s' , $N_{css't}$ is the number of individuals living in state s at time t who were born in state s' , and θ_s and θ_t are state and year fixed effects. Required years of schooling correspond to the time a children born in a given year is required to stay in school, calculated by combining information on required attendance at each year of life (see [Clay, Lingwall, and Stephens, 2021](#); [Stephens and Yang, 2014](#)). The instrument is thus average *required* years of schooling of the working-age population, calculated by averaging required years across all cohort-state-of-birth cells, weighted by their populations. This approach is analogous to the one recently adopted by [Guo, Roys, and Seshadri \(2018\)](#).

Results Table C5 presents the main results. In the baseline specification, the aggregate return to schooling is 0.16 log points. The corresponding values are 0.44 for the bottom 20% and 0.05 for the top 20%. Education thus appears as a powerful driver of inequality reduction, even more so in the U.S. than in India and Indonesia. Adding interacted census region and year fixed effects leaves the results almost unchanged (columns 4 to 6). Columns 7 to 9 further add controls for initial conditions. The aggregate effect is slightly lower and the estimates are unsurprisingly underpowered. Even under this highly demanding specification, however, the coefficient on the bottom 20% remains large and statistically significant.

Table C6 compares observed and simulated effects of the policy. The simulation is done by upgrading the education of randomly sampled individuals with either no schooling or primary education to secondary education, given that required years of schooling range from 0 to 9 years. Here, the model appears to strongly underestimate the aggregate and inequality-reducing effects of the policy, even with a return to schooling of 16% and an elasticity of substitution of 2.

There are at least three reasons that could explain this finding. First, state compulsory schooling laws extended both primary and secondary school attendance, with significant variations in timing and intensity across states. This makes it more difficult to accurately simulate the overall effect of these policies. Second, there is evidence that returns to schooling were substantially higher at the bottom of the income distribution during the first wave of compulsory schooling laws (Clay, Lingwall, and Stephens, 2021). The simulation assumes a constant return by income group, which by construction limits its ability to capture higher returns for low-income earners. Third, recent evidence points to large human capital externalities from schooling expansion in the United States (Guo, Roys, and Seshadri, 2018). This might explain why the simulation ends up strongly underestimating the aggregate return to schooling in this context.

Table C1 – India DPEP: Aggregate and Distributional Effects of Schooling

	Bandwidth Selection: CCT Method			Bandwidth Selection: I and K Method		
	Average Income	Bottom 20% Income	Top 20% Income	Average Income	Bottom 20% Income	Top 20% Income
Average Years of Schooling	0.117* (0.061)	0.207*** (0.059)	-0.092 (0.071)	0.257*** (0.058)	0.316*** (0.058)	0.012 (0.068)
N	46314	9007	9515	46314	9007	9515

Notes. The table reports the effect of district average years of schooling on district average income, the average of the bottom 20%, and the average income of the top 20%. Bandwidths: “CCT” indicates the [Calonico, Cattaneo, and Titiunik \(2014\)](#) method, “I and K” the [Imbens and Kalyanaraman \(2012\)](#) method. Data from [Khanna \(2023\)](#). Standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C2 – India DPEP: Actual vs. Simulated Effects of Educational Expansion

	Parameters		Effect of Increasing Average District Schooling by One Year (%)					
	Return to Schooling	Elasticity of Substitution	Average Income	Q1	Q2	Q3	Q4	Q5
Actual Effect			12.4	23.0	11.9	18.7	21.3	-8.8
Simulated Effect	13%	∞	9.3	17.5	13.5	14.9	10.0	5.6
	13%	6	9.3	19.3	13.4	14.9	10.0	5.5
	13%	4	9.3	20.2	13.4	14.8	10.0	5.4
	13%	2	9.3	22.7	12.9	14.9	9.9	5.2
	16%	∞	12.4	19.0	15.1	18.6	14.4	8.4
	16%	6	12.4	21.0	15.1	18.7	14.3	8.1
	16%	4	12.4	22.0	15.0	18.7	14.3	8.0
	16%	2	12.4	25.0	14.8	19.0	14.3	7.7
	20%	∞	17.1	20.1	16.6	22.1	21.2	13.5
	20%	6	17.1	22.4	16.9	22.4	21.1	13.1
	20%	4	17.1	23.6	17.0	22.5	21.1	12.9
	20%	2	17.1	27.0	17.1	23.1	20.9	12.4

Notes. Actual effect: estimated effect of the policy on average district income and the average income of each wage quintile, using data from [Khanna \(2023\)](#). Simulated effect: effect of the policy predicted using 2019 LFS data, under different assumptions on the return to a year of schooling and the elasticity of substitution between skilled and unskilled workers.

Table C3 – Indonesia INPRES: Aggregate and Distributional Effects of Schooling

	Baseline			+ Controlling for 1971 Primary School Enrollment and Water & Sanitation Spending			+ Controlling for 1971 Child Population and Population Density		
	Average Income	Bottom 20% Income	Top 20% Income	Average Income	Bottom 20% Income	Top 20% Income	Average Income	Bottom 20% Income	Top 20% Income
Average Years of Schooling	0.087*** (0.026)	0.220*** (0.035)	0.058* (0.031)	0.133 (0.083)	0.505*** (0.145)	-0.002 (0.097)	0.084 (0.108)	0.445** (0.179)	-0.029 (0.131)
First-Stage F-Stat	174.56	174.56	174.56	18.32	18.32	18.32	10.53	10.53	10.53
N	5520	5520	5520	5352	5352	5352	5304	5304	5304

Notes. The table reports the effect of regency average years of schooling on regency average income, the average income of the bottom 20%, and the average income of the top 20%. Columns 1 to 3 control for the demographic composition of the regency, the share of women, and the share of workers with tertiary education. Columns 4 to 6 add controls for 1971 primary school enrollment rates and water and sanitation spending, interacted with survey year. Columns 7 to 9 further add controls for the share of the population aged 5 or below in 1971 and population density in 1971, interacted with survey year. Data from [Duflo \(2001\)](#) and [Roodman \(2022\)](#). Standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C4 – Indonesia INPRES: Actual vs. Simulated Effects of Educational Expansion

	Parameters		Effect of Increasing Average District Schooling by One Year (%)					
	Return to Schooling	Elasticity of Substitution	Average Income	Q1	Q2	Q3	Q4	Q5
Actual Effect			9.1	24.6	16.7	12.2	8.4	6.0
Simulated Effect	9%	∞	5.7	15.4	10.7	5.3	4.6	4.0
	9%	6	5.7	19.2	11.7	6.4	5.2	3.7
	9%	4	5.7	20.5	12.3	7.1	5.5	3.5
	9%	2	5.7	25.6	13.9	4.4	3.2	3.0
	11%	∞	7.7	17.3	12.9	7.7	7.3	5.7
	11%	6	7.7	20.7	13.9	7.5	7.1	5.4
	11%	4	7.7	22.6	14.6	8.6	7.8	5.3
	11%	2	7.7	28.2	16.5	6.0	5.8	5.1
	13%	∞	10.5	18.9	15.2	9.6	10.1	8.9
	13%	6	10.5	22.2	15.9	9.3	9.8	8.6
	13%	4	10.5	24.1	16.3	9.0	9.7	8.6
	13%	2	10.5	29.1	18.7	7.6	8.2	8.2

Notes. Actual effect: estimated effect of the policy on average district income and the average income of each wage quintile, using data from [Duflo \(2001\)](#). Simulated effect: effect of the policy predicted using 1996 SAKERNAS microdata, under different assumptions on the return to a year of schooling and the elasticity of substitution between skilled and unskilled workers.

Table C5 – U.S. Compulsory Schooling Laws: Aggregate and Distributional Effects of Schooling

	Baseline			+ Census Region × Year FE			+ Controls for 1940 Educational Attainment and Average Income × Year FE		
	Average Income	Bottom 20% Income	Top 20% Income	Average Income	Bottom 20% Income	Top 20% Income	Average Income	Bottom 20% Income	Top 20% Income
Average Years of Schooling	0.157*** (0.032)	0.437*** (0.095)	0.050* (0.027)	0.147*** (0.045)	0.431*** (0.110)	0.079* (0.044)	0.082 (0.051)	0.272** (0.114)	0.063 (0.057)
First-Stage F-Stat	87.05	87.05	87.05	19.01	19.01	19.01	12.10	12.10	12.10
N	343	343	343	343	343	343	343	343	343

Notes. The table reports the effect of state average years of schooling on state average income, the average income of the bottom 20%, and the average income of the top 20%. Columns 1 to 3 control for the demographic, gender, and racial composition of each state, as well as the share of workers with tertiary education. Columns 4 to 6 add census region × year fixed effects. Columns 7 to 9 further add controls for 1940 average years of schooling and average personal income, interacted with survey year dummies. Data from IPUMS census microdata combined with information on compulsory schooling laws from [Acemoglu and Angrist \(2000\)](#) and [Clay, Lingwall, and Stephens \(2021\)](#). Standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C6 – U.S. Compulsory Schooling Laws: Actual vs. Simulated Effects of Educational Expansion

	Parameters		Effect of Increasing Average State Schooling by One Year (%)					
	Return to Schooling	Elasticity of Substitution	Average Income	Q1	Q2	Q3	Q4	Q5
Actual Effect			16.9	54.9	58.4	29.6	13.1	5.1
Simulated Effect	8%	∞	4.5	17.5	9.7	5.2	3.6	2.8
	8%	6	4.5	21.1	11.4	5.7	3.3	2.1
	8%	4	4.5	22.8	12.3	6.0	3.2	1.8
	8%	2	4.5	28.0	14.9	6.8	2.8	0.9
	12%	∞	7.4	24.2	14.1	7.6	5.5	6.0
	12%	6	7.4	27.7	15.8	8.2	5.3	5.3
	12%	4	7.4	29.5	16.7	8.5	5.2	4.9
	12%	2	7.4	34.8	19.2	9.3	4.7	4.0
	16%	∞	11.0	29.5	18.4	9.7	7.5	10.6
	16%	6	11.0	33.1	20.2	10.3	7.2	9.9
	16%	4	11.0	34.9	21.1	10.6	7.1	9.6
	16%	2	11.0	40.2	23.6	11.4	6.7	8.6

Notes. Actual effect: estimated effect of the policy on average state income and the average income of each personal income quintile, combining IPUMS census microdata with information on compulsory schooling laws from [Acemoglu and Angrist \(2000\)](#) and [Clay, Lingwall, and Stephens \(2021\)](#). Simulated effect: effect of the policy predicted using 1960 census microdata, under different assumptions on the return to a year of schooling and the elasticity of substitution between skilled and unskilled workers.

D. Robustness Checks and Sensitivity Analysis

D.1. Causal Estimates of the Returns to Schooling

A New Compilation of IV Returns to Schooling In an alternative specification, I rely on instrumental variable estimates of the returns to schooling collected from a number of existing studies. Surveying the literature, I was able to find 62 studies exploiting natural experiments to estimate the causal effect of education on earnings. Table [D1](#) provides a complete list of these articles, together with the type of policy, the corresponding education level, and the estimated coefficients and standard errors when available.

This article is interested in the effect of educational attainment on earnings. I thus start by excluding studies that do not allow estimating a Mincerian return to completed years of schooling. This is the case of 15 studies that focused on compulsory schooling laws in France, Germany, the Netherlands, and the United Kingdom. These studies have one thing in common: either the policy studied did not lead to an increase in educational attainment, or the data used by the authors did not cover completed years of schooling. In the United Kingdom, for instance, ten studies estimated returns to compulsory schooling using the school leaving age as the explanatory variable. Yet the leaving age can differ significantly from educational attainment, as documented for instance by [Grenet \(2013\)](#) in the case of France’s 1967 Berthoin reform, which increased the age at which French students left school but had no effect on attainment. As a result, it is unclear whether zero returns found in some of these papers reflect the failure of the policy to raise attainment or the actual absence of a return to educational attainment. For this reason, I exclude these studies focusing on UK reforms, as well as the two French and UK reforms studied by [Grenet \(2013\)](#), who also reports returns as a function of the leaving age but not as a function of completed years of schooling. Similarly, [Pischke \(2007\)](#) finds no effect of a reform lengthening the school year on attainment in Germany, while the data used in [Pischke and Wachter \(2008\)](#) do not allow them to study the effect of German compulsory schooling on educational attainment (their first stage is based on the secondary school graduation year). [Oosterbeek and Webbink \(2007\)](#) study the effect of a compulsory schooling law in the Netherlands, but do not observe educational attainment in their dataset either. I also exclude studies that do not have sufficient statistical power to rule out either zero or large returns ([Duflo, Dupas, and Kremer, 2024](#); [Filmer and Schady, 2014](#)).

This leaves us with a collection of 46 studies. Among these, 13 report an IV estimate of the return to schooling, but do not report an OLS estimate for comparison. My comparison of OLS and IV returns thus ends up covering 33 separate studies.

Discussion IV estimates of the returns to schooling have two key advantages: they are causally identified, and they focus on newly skilled workers. Exploiting the results of these studies in my analysis is not without difficulties, however, for at least four reasons.

First, these studies cover specific populations, which marginally gained access to education as a result of each policy. These populations may be different from the average newly skilled worker in a given country.

Second, each study covers a specific education level. This implies that the IV return reflects the return to gaining access to a specific level, while the OLS return reflects the average return across all levels. Depending on the convexity of returns, the gap between the degree-specific and average return can be large. In particular, most studies covered in this collection expanded access to basic education. In the presence of convex returns, this would imply that the IV-OLS gap is underestimated.

Third, studies estimate the IV return to schooling among workers falling into specific age ranges. Treated groups are younger than the average worker in many studies. In the presence of growing education premiums over the life-cycle, this implies that IV returns may substantially underestimate the lifetime economic benefits of schooling (Bhuller, Mogstad, and Salvanes, 2017).

Fourth, education policies may have spillovers over non-treated groups due to general equilibrium effects. In the presence of imperfect substitutability across skill and age groups, the typical IV estimate of the return to schooling will be biased (Khanna, 2023). Given large variations in the scale of education policies across articles, it is hard to conclude on the significance of these general equilibrium effects.

Bringing the Model to the Microdata with IV Returns to Schooling With these limitations in mind, I investigate the sensitivity of my results to using IV returns to schooling instead of Mincerian returns estimated by OLS. I start by exploiting the ratio of IV to OLS returns to schooling estimated in each study to correct OLS returns to schooling estimated with my data. Let r_{cj}^{OLS} be the OLS return to schooling estimated for country c and education level j , r_{cj}^{IV} the IV return estimated in the same paper, and $\hat{r}_{cj}^{OLS}$ the OLS return estimated with my microdatabase. I construct “IV-corrected” returns to schooling as:

$$\hat{r}_{cj}^{IV} = \hat{r}_{cj}^{OLS} \times \frac{r_{cj}^{IV}}{r_{cj}^{OLS}} \quad (D1)$$

Hence, I correct OLS returns estimated with my data using country-education-level-specific adjustment factors derived from the comparison of OLS and IV returns found in each paper. This allows me to adjust returns to schooling in 23 countries representative of about two-thirds of the world’s population.

I then adjust OLS returns to schooling in missing countries. For European countries not covered by any study, I use the 1.33 correction factor found in [Brunello, Weber, and Weiss \(2015\)](#), who estimate the returns to schooling in Europe by combining information on various European compulsory schooling laws. For other countries, I use the average correction factor by education level found across studies. This amounts to multiplying the returns to primary, secondary, and tertiary education by about 1.4, 1.3, and 2, respectively.

Finally, I construct the distributional growth accounting decomposition using these returns. First, I downgrade education levels in each survey until reaching the 1980 counterfactual, as in the rest of the paper. Second, I reduce earnings of “treated” individuals using these “IV-corrected” returns to schooling. Third, I aggregate actual and counterfactual income by decile. Finally, I adjust relative counterfactual incomes by decile so that they match relative incomes predicted by the model. In other words, the only difference between this specification and my main results is that the *aggregate* effects of education in each country are captured by the “IV-corrected” returns rather than by wage adjustments predicted by the model. Distributional effects of schooling are kept identical in the two specifications.

D.2. Alternative Elasticities and Production Functions

CES Production Function With Alternative Nesting Structure The production function is:

$$H = \left(A_{low} L_{low}^{\frac{\sigma_1-1}{\sigma_1}} + A_{upp} L_{upp}^{\frac{\sigma_1-1}{\sigma_1}} \right)^{\frac{\sigma_1}{\sigma_1-1}} \quad (D2)$$

$$L_{low} = \left(A_{non} L_{non}^{\frac{\sigma_2-1}{\sigma_2}} + A_{pri} L_{pri}^{\frac{\sigma_2-1}{\sigma_2}} \right)^{\frac{\sigma_2}{\sigma_2-1}} \quad (D3)$$

$$L_{upp} = \left(A_{sec} L_{sec}^{\frac{\sigma_3-1}{\sigma_3}} + A_{ter} L_{ter}^{\frac{\sigma_3-1}{\sigma_3}} \right)^{\frac{\sigma_3}{\sigma_3-1}} \quad (D4)$$

Returns to primary, secondary, and tertiary education are:

$$r_{pri} = \ln\left(\frac{w_{pri}}{w_{non}}\right) = \ln\left(\frac{A_{pri}}{A_{non}}\right) - \frac{1}{\sigma_2} \ln\left(\frac{L_{pri}}{L_{non}}\right) \quad (D5)$$

$$r_{sec} = \ln\left(\frac{w_{sec}}{w_{pri}}\right) = \ln\left(\frac{A_{upp}}{A_{low}}\right) + \ln\left(\frac{A_{sec}}{A_{pri}}\right) - \frac{1}{\sigma_1} \ln\left(\frac{L_{upp}}{L_{low}}\right) - \frac{1}{\sigma_3} \ln\left(\frac{L_{sec}}{L_{upp}}\right) - \frac{1}{\sigma_2} \ln\left(\frac{L_{low}}{L_{pri}}\right) \quad (D6)$$

$$r_{ter} = \ln\left(\frac{w_{ter}}{w_{sec}}\right) = \ln\left(\frac{A_{ter}}{A_{sec}}\right) - \frac{1}{\sigma_3} \ln\left(\frac{L_{ter}}{L_{sec}}\right) \quad (D7)$$

Let $A_{upp} = 1 - A_{low}$, $A_{pri} = 1 - A_{non}$ and $A_{ter} = 1 - A_{sec}$. The model can then be solved in four steps.

First, recover technology parameters within each nest:

$$A_{non} = \frac{1}{1 + \exp\left(r_{pri} + \frac{1}{\sigma_2} \ln\left(\frac{L_{pri}}{L_{non}}\right)\right)} \quad (D8)$$

$$A_{pri} = 1 - A_{non} \quad (D9)$$

$$A_{sec} = \frac{1}{1 + \exp\left(r_{sec} + \frac{1}{\sigma_3} \ln\left(\frac{L_{ter}}{L_{sec}}\right)\right)} \quad (D10)$$

$$A_{ter} = 1 - A_{sec} \quad (D11)$$

Second, use these parameters to construct L_{low} and L_{upp} . Third, recover A_{low} and A_{upp} :

$$A_{low} = \frac{1}{1 + \exp\left(r_{sec} - \ln\left(\frac{A_{sec}}{A_{pri}}\right) + \frac{1}{\sigma_1} \ln\left(\frac{L_{upp}}{L_{low}}\right) + \frac{1}{\sigma_3} \ln\left(\frac{L_{sec}}{L_{upp}}\right) + \frac{1}{\sigma_2} \ln\left(\frac{L_{low}}{L_{pri}}\right)\right)} \quad (D12)$$

$$A_{upp} = 1 - A_{low} \quad (D13)$$

Fourth, use these parameters to construct H .

CES Production Function With No Nest I also consider a CES production function with no nest:

$$H = \left(A_{non} L_{non}^{\frac{\sigma-1}{\sigma}} + A_{pri} L_{pri}^{\frac{\sigma-1}{\sigma}} + A_{sec} L_{sec}^{\frac{\sigma-1}{\sigma}} + A_{ter} L_{ter}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

Returns to primary, secondary, and tertiary education are:

$$\begin{aligned} r_{pri} &= \ln\left(\frac{w_{pri}}{w_{non}}\right) = \ln\left(\frac{A_{pri}}{A_{non}}\right) - \frac{1}{\sigma} \ln\left(\frac{L_{pri}}{L_{non}}\right) \\ r_{sec} &= \ln\left(\frac{w_{sec}}{w_{pri}}\right) = \ln\left(\frac{A_{sec}}{A_{pri}}\right) - \frac{1}{\sigma} \ln\left(\frac{L_{sec}}{L_{pri}}\right) \\ r_{ter} &= \ln\left(\frac{w_{ter}}{w_{sec}}\right) = \ln\left(\frac{A_{ter}}{A_{sec}}\right) - \frac{1}{\sigma} \ln\left(\frac{L_{ter}}{L_{sec}}\right) \end{aligned}$$

The model can be estimated in two steps. First, recover technology parameters, normalizing A_{non} to 1:

$$\begin{aligned} A_{non} &= 1 \\ A_{pri} &= \exp\left(r_{pri} + \frac{1}{\sigma} \ln\left(\frac{L_{pri}}{L_{non}}\right)\right) \\ A_{sec} &= \exp\left(\ln(A_{pri}) + r_{sec} + \frac{1}{\sigma} \ln\left(\frac{L_{sec}}{L_{pri}}\right)\right) \\ A_{ter} &= \exp\left(\ln(A_{sec}) + r_{ter} + \frac{1}{\sigma} \ln\left(\frac{L_{ter}}{L_{sec}}\right)\right) \end{aligned}$$

Second, construct H using these parameters.

CES Production Function With Imperfect Substitution Between Age Groups I also consider an extension in which workers belonging to different age groups are imperfectly substitutable. For simplicity, I consider a single elasticity across skill groups σ and a single elasticity across age groups λ . The production function is:

$$H = \left(A_{non} L_{non}^{\frac{\sigma-1}{\sigma}} + A_{pri} L_{pri}^{\frac{\sigma-1}{\sigma}} + A_{sec} L_{sec}^{\frac{\sigma-1}{\sigma}} + A_{ter} L_{ter}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (D14)$$

Each skill group i is composed of workers belonging to different age groups:

$$L_i = \left(A_{ia} L_{ia}^{\frac{\lambda-1}{\lambda}} + A_{ib} L_{ib}^{\frac{\lambda-1}{\lambda}} + A_{ic} L_{ic}^{\frac{\lambda-1}{\lambda}} + A_{id} L_{id}^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} \quad (D15)$$

Where a , b , c , and d refer to workers aged 25 to 34, 35 to 44, 45 to 54, and 55 and above, respectively. It is straightforward to solve this model using the same steps as in the previous sections. The wage of a worker belonging to skill group i and age group j is given by:

$$W_{ij} = H^{\frac{1}{\sigma}} A_i L_i^{-\frac{1}{\sigma}} L_i^{\frac{1}{\lambda}} A_{ij} L_{ij}^{-\frac{1}{\lambda}} \quad (D16)$$

The return to experience for group j compared to group j' is:

$$r_{jj'} = \ln\left(\frac{w_{ij}}{w_{ij'}}\right) = \ln\left(\frac{A_{ij}}{A_{ij'}}\right) - \frac{1}{\lambda} \ln\left(\frac{L_{ij}}{L_{ij'}}\right) \quad (D17)$$

I solve this model and bring it to the microdata in six steps.

First, I run OLS regressions relating the log of personal income to dummies for belonging to these different age groups, controlling for education and gender. This allows me to recover estimates of the

returns to experience $r_{jj'}$ for the different age groups in each country.

Second, I recover age-specific productivity parameters A_{ij} from equation D17. To measure L_{ij} in each country, I rely on the Barro-Lee database, which records the number of workers belonging to different skill and age groups in each country. I assume an elasticity of substitution across age groups of $\lambda = 5$ (e.g., Card and Lemieux, 2001; Fernández and Messina, 2018; Khanna, 2023).

Third, once A_{ij} are known, I construct L_i for the different skill groups.

Fourth, I recover technology parameters A_i for the different skill groups, using estimates of the returns to schooling as in the rest of the paper.

Fifth, I construct actual output and counterfactual output absent educational expansion and demographic change, replacing labor supplies by their 1980 counterparts. Notice that I replace the entire distribution of workers by skill and age group by its 1980 value. This counterfactual thus incorporates both returns to education and returns to experience.

Sixth, I bring the model to the microdata, using the same methodology as in the rest of the paper. The main difference here is that workers see both their education and their age change in the counterfactual. Their wage thus ends up affected by both skill supply and age supply effects, which are governed by these two elasticities. The outcome of this exercise is a counterfactual distribution of income absent both educational expansion and demographic change.

D.3. Within-Category Heterogeneity in Human Capital

My main specification assumes that workers are homogeneous within each of the four skill groups (no schooling, primary, secondary, and tertiary). In this section, I consider an extension of the model in which human capital is allowed to vary within each education category.

Model My analysis follows Caselli (2016). Workers are imperfectly substitutable across the four groups, while workers within each category are perfectly substitutable but may still have different human capital levels. More specifically, let us rewrite equations 15 to 17:

$$H = \left(A_{\overline{ter}} L_{\overline{ter}}^{\frac{\sigma_1-1}{\sigma_1}} + A_{ter} L_{ter}^{\frac{\sigma_1-1}{\sigma_1}} \right)^{\frac{\sigma_1}{\sigma_1-1}} \quad (D18)$$

$$L_{\overline{ter}} = \left(A_{\overline{sec}} L_{\overline{sec}}^{\frac{\sigma_2-1}{\sigma_2}} + A_{sec} L_{sec}^{\frac{\sigma_2-1}{\sigma_2}} \right)^{\frac{\sigma_2}{\sigma_2-1}} \quad (D19)$$

$$L_{\overline{sec}} = \left(A_{non} L_{non}^{\frac{\sigma_3-1}{\sigma_3}} + A_{pri} L_{pri}^{\frac{\sigma_3-1}{\sigma_3}} \right)^{\frac{\sigma_3}{\sigma_3-1}} \quad (D20)$$

My benchmark specification assumes $L_{pri} = L_{pri1} + L_{pri2}$ and $L_{sec} = L_{sec1} + L_{sec2}$, where L_{pri1} , L_{pri2} , L_{sec1} , and L_{sec2} refer to the share of workers with incomplete primary education, complete primary education, incomplete secondary education, and complete secondary education, respectively. I now consider an extension in which these groups are allowed to hold different human capital, as in [Caselli \(2016\)](#):

$$L_{pri} = L_{pri1} + e^{r_{pri2}} L_{pri2} \quad (D21)$$

$$L_{sec} = L_{sec1} + e^{r_{sec2}} L_{sec2} \quad (D22)$$

Where r_{pri2} and r_{sec2} are the returns to completing primary and secondary education with respect to holding incomplete degrees, respectively. In the absence of data on incomplete versus complete tertiary education, I continue to assume that workers with tertiary education are homogeneous. My main results correspond to the case where $r_{pri2} = r_{sec2} = 0$: increasing the share of workers with complete relative to incomplete secondary education has no effect on output. In this extended model, in contrast, doing so increases output as long as the return to completing secondary education is positive.

Empirical Application I bring this model to the data using surveys covering sufficiently detailed information on educational attainment. Unfortunately, the ILO microdata do not provide information on incomplete versus complete degrees. For this, I turn to the I2D2 database. I searched for all surveys fielded after 2015 that cover the required information. I was able to identify such surveys for 79 countries. Of these, only 37 have sufficiently large sample sizes to obtain reliable estimates of the returns to completing primary and secondary education versus holding incomplete degrees. My analysis therefore focuses on these countries.

I estimate returns to schooling using the following modified Mincerian equation:

$$\ln y_i = \alpha + \beta_{pri1} D_{i,pri1} + \beta_{pri2} D_{i,pri2} + \beta_{sec1} D_{i,sec1} + \beta_{sec2} D_{i,sec2} + \beta_{ter} D_{i,ter} + X_i \beta + \varepsilon_i \quad (D23)$$

Where y_i is total annual earned income from all jobs of individual i in a given country, $D_{i,pri1}$, $D_{i,pri2}$, $D_{i,sec1}$, $D_{i,sec2}$, and $D_{i,ter}$ are dummies for having reached incomplete primary, complete primary, incomplete secondary, complete secondary, and tertiary education, and X_i is a vector of controls including gender, an age quartic, and interactions between gender and the age quartic.

Returns to completing primary and secondary education are:

$$r_{pri2} = \beta_{pri2} - \beta_{pri1} \quad (D24)$$

$$r_{sec2} = \beta_{sec2} - \beta_{sec1} \quad (D25)$$

All other ingredients of the model stick to the benchmark specification. In particular, technology terms can be recovered from the same estimates of returns to primary, secondary, and tertiary education as in the rest of the paper. The share of workers with incomplete versus complete degrees is taken from the Barro-Lee database. All parameters are then known and one can estimate counterfactual wages and output.

Results Figure D1 compares schooling gains estimated with the benchmark versus extended models. Table D5 reports detailed results by country. In the average country, the return to completing primary education versus not completing it is 14%, while the return to completing secondary education is 27%. Annualized schooling gains are 1% in the benchmark model versus 1.1% in the extended model. The share of growth explained by education is 39% versus 43%. As shown in Figure D1, all countries are very close to the 45-degree line: accounting for within-category variations in human capital barely affects cross-country differences in schooling gains.

If anything, the extended model predicts slightly larger effects of educational expansion. The reason for this is that the relative shares of workers with complete degrees has increased. In the average country, 47% of primary-educated workers had complete degrees in 1980, compared to 74% in 2019. 26% of secondary-educated workers had complete degrees in 1980, compared to 54% in 2019. My benchmark specification treats these changes as having no implication for economic growth, while the extended model does account for the resulting positive returns. The returns to completing primary and secondary education are low, however, which is why this refinement barely affects estimates of the overall contribution of education.

D.4. Heterogeneous Educational Expansion by Socioeconomic Characteristic

In my main specification, I randomly sample individuals and downgrade their education by age-gender cell. In this section, I investigate the implications of further refining the analysis by accounting for heterogeneous educational expansion by socioeconomic characteristic other than age and gender.

I focus on three countries with available data: India, South Africa, and the United States. I compare three specifications in each country. The first one downgrades education levels of individuals without any

heterogeneity, the second one accounts for heterogeneous educational expansion by age and gender, and the third one further accounts for differences in educational progress by region (India) or region and race (South Africa, United States). The earnings of “treated” individuals are then reduced using estimates of the returns to schooling by level, as in the main analysis.

For India, the distribution of educational attainment by age, gender, and state of residence in 1983 is estimated using the 1983 National Sample Survey. The simulation is then run on the 2019 Periodic Labor Force Survey. For South Africa, educational attainment by age, gender, race, and province is estimated using the 2002 General Household Survey. The simulation is then run on the 2019 General Household Survey. For the United States, educational attainment by age, gender, state, and race is estimated using the 1980 Current Population Survey. The simulation is then run on the 2019 Current Population Survey.

Figure D2 plots schooling gains by income quintile, defined as the percent difference between actual income and counterfactual income absent educational expansion. As shown in the Figure, accounting for heterogeneous educational expansion by socioeconomic characteristic only marginally affects estimates of the aggregate and distributional effects of education.

D.5. Lost Income Due to School Attendance

If education had not improved, fewer young adults would be in school today. These individuals could be working instead of studying. In this section, I attempt to derive an upper bound on this opportunity cost. I consider an extreme scenario in which all individuals aged 15-25 currently attending school would instead be on the labor market if education had not improved. I assume that in this counterfactual, these individuals would face the same employment rate as other individuals in the same age group. The additional share of the adult population that would be employed in the counterfactual is:

$$\begin{aligned} \frac{\text{Newly Employed}}{\text{Adult Population}} &= \text{Share of Adults Aged 15-25} \\ &\quad \times \text{School Attendance Rate Among Adults Aged 15-25} \\ &\quad \times \text{Employment Rate Among Out-of-School Adults Aged 15-25} \end{aligned} \tag{D26}$$

In the average country, 26% of the adult population was aged 15-25 in 2019, and 44% of them were in school. Among those not in school, 55% were employed. This means that if all individuals aged 15-25 were to join the labor market instead of going to school, the share of the adult population in employment would

increase by $0.26 \times 0.44 \times 0.55 = 0.06$. 6% of adults would thus start working.

I then assume that these additional workers would receive the same wage as workers aged 15-25 who are currently employed. Lost income as a share of total labor income is thus:

$$\frac{\text{Lost Income}}{\text{Total Labor Income}} = \frac{\text{Newly Employed}}{\text{Adult Population}} \times \frac{\text{Average Labor Income of Workers Aged 15-25}}{\text{Average Labor Income}} \quad (\text{D27})$$

In the average country, the average income of workers aged 15-25 is about two-thirds of that of the overall adult population, so the opportunity cost of schooling amounts to about 4% of total labor income. To quantify the implications of this opportunity cost, I simply add this lost income to counterfactual incomes $y^P(A^{2019}, L^{1980})$ in each country. In other words, counterfactual income absent educational expansion would be higher in this scenario because of additional income generated by newly employed workers.

With these assumptions, I find that education accounts for 41% of global economic growth and 52% of growth among the world's poorest 20%. This should be seen as a very conservative lower bound, given that many young adults would still be in school today even if education had not improved since 1980.

D.6. Education Quality

Changes in education quality could affect the results of this paper. If education quality has increased or decreased, then educational attainment becomes a biased measure of actual changes in the education of the labor force: 1980 and 2019 levels of attainment are not comparable indicators anymore. In this section, I investigate available data on the evolution of education quality and implications for my analysis.

D.6.1. Trends in Education Quality: A Comparison of Available Estimates

International Test Scores Drawing from various international sources, [Angrist et al. \(2021\)](#) compile test score results for 163 countries over the 2000-2017 period, 122 of which have at least one data point in the 2000s and another data point in the 2010s. Figure [D3a](#) compares average test scores in the 2000s and 2010s for all countries with available data. Each data point corresponds to a test score in a given country, for a given education level (primary/secondary) and subject (mathematics/science/reading). All points are very close to the 45-degree line, suggesting that there has been little change in quality over the period. If anything, there has been a slight improvement in average quality: test scores have improved for 170 country-level-subject cells, while they have declined for 100.

For a more restricted number of countries, it is also possible to look at longer-run trends in education

quality, based on the database compiled by [Altinok, Angrist, and Patrinos \(2018\)](#). Figure D3b plots the evolution of this indicator since 1970 for a selected number of high- and middle-income countries. The picture that arises is again one of remarkable stability, although some countries have undergone important long-run improvements, including Brazil, Chile, Iran, and South Korea.

Conditional Literacy Test scores suffer from a critical lack of historical depth for most countries in the world. To make a first step towards closing this gap, [Le Nestour, Moscoviz, and Sandefur \(2022\)](#) exploit information on literacy reported in the Demographic and Health Surveys and the Multiple Indicator Cluster Surveys. Based on this, [Le Nestour, Moscoviz, and Sandefur \(2022\)](#) exploit repeated cross-sections to identify changes in education quality by cohort, defined as expected literacy at grade 5. Figure D3c shows the main result of this exercise, comparing expected literacy for cohorts born in 1950-1960s versus 1980-2000. The estimates of [Le Nestour, Moscoviz, and Sandefur \(2022\)](#) point to a clear decline in quality in a number of developing countries.

These results are insightful, but it is important to stress that they do not necessarily imply that the results presented in this paper should be revised downwards for at least four reasons.

First, ability to read is a partial measure of quality. For instance, [Hermo et al. \(2022\)](#) show that the decline of vocabulary knowledge in Sweden since the 1960s has been accompanied by a significant increase in logical reasoning skills, which can be rationalized by increasing labor market returns to the latter. Second, identifying trends in the quality of education from repeated cross sections of surveys requires explicitly modeling age, period, and cohort effects, which makes the results more sensitive to methodological choices and measurement error. Third, such estimates are not immune to standard problems associated with causal identification (which is also true of test scores). Finally, changes in average performance may not necessarily imply lower returns to schooling. Even if newly educated cohorts may have lower levels of cognitive skills, economic returns to schooling for them may still be equal, or even greater (as suggested by the IV returns to schooling presented in the main text), than returns for the rest of the population. Put differently, differences in average skills may be very different from differences in marginal returns to skill.

Returns to Schooling Among U.S. Migrants A last piece of evidence comes from returns to schooling among U.S. migrants. [Schoellman \(2012\)](#) argues that differences in returns to schooling among U.S. migrants originating from different countries provides a good proxy for education quality, because it captures income gains from schooling for individuals having been educated in different countries but working in the same

labor market. [Schoellman \(2012\)](#) provides evidence that this indicator is a good proxy for education quality, strongly correlating with GDP per capita and available test scores (see also [Rossi, 2022](#)).

I investigate trends in returns to schooling among U.S. migrants by pooling the 1980, 1990, and 2000 U.S. censuses, together with all American Community Surveys from 2001 to 2021. I restrict the sample to individuals aged 25 to 65 with positive earned income, who were born outside of the U.S. between 1950 and 1980, and arrived in the U.S. after age 20. I run the following regressions:

$$y_{icyt} = \zeta_{cy}s_{icyt} + X_{icyt}\beta_{cy} + \mu_t + \varepsilon_{icyt} \quad (\text{D28})$$

With y_{icyt} the log of total annual earned income of individual i born in country c in decade y (1950s, 1960s, 1970s, or 1980s) and observed in year t . s_{icyt} is completed years of schooling, X_{icyt} are control variables (gender, state of residence, and year of immigration), and μ_t are census/ACS year fixed effects. The parameter of interest is ζ_{cy} , the return to a year of schooling for individuals born in country c in decade y . If education quality has declined substantially, then we should expect ζ_{cy} to have declined over time: a year of schooling should deliver greater returns for migrants born in the 1950s than for migrants born in the 1980s. I run this regression separately for each country of origin $\times$ decade of birth cell.

Figure [D3d](#) plots population-weighted averages of the estimated returns to schooling by world region of birth and decade of birth. Returns to schooling are lowest among migrants from Latin America and Sub-Saharan Africa and highest among migrants from Europe and the Anglosphere. There are fluctuations across decades, but no clear trend in quality in most regions. The world average varies from 7% to 9% with no clear trend.⁵ This suggests again that changes in education quality are unlikely to play a substantial role in affecting the results presented in this paper.

D.6.2. A Quantification Exercise

While it remains unclear which data source should be preferred, it is still useful to test how sensitive my main findings are to accounting for the decline in quality documented in [Le Nestour, Moscoviz, and Sandefur \(2022\)](#). This is somewhat of a heroic task, because it requires (1) extrapolating cohort trends to cover education quality for the entire 1980-2019 working-age populations (2) putting a monetary value on literacy to build measures of quality-adjusted years of schooling, and (3) extrapolating changes in quality to

⁵Interestingly, the raw cross-country correlation between changes in these returns and changes in the [Le Nestour, Moscoviz, and Sandefur \(2022\)](#) indicator from the 1960s to 1980s cohorts is 0.33. This suggests that the two sources tell a broadly similar story on which countries have seen education quality decline or improve most.

countries with no available data. This section represents an exploratory attempt at doing so.

Methodological Framework Consider the standard extension of the Mincer-type human capital stock (e.g., [Hanushek, Ruhose, and Woessman, 2017](#)):

$$h = \exp(r_L L + r_Q Q) \quad (\text{D29})$$

With r_L the return to a year of schooling, L average years of schooling, r_Q the return to education quality, and Q an indicator of education quality. The objective is to convert a change in quality from Q to $\tilde{Q}$ into an equivalent change in years of schooling from L to $\tilde{L}$. This equivalence satisfies:

$$\exp(r_L L + r_Q \tilde{Q}) = \exp(r_L \tilde{L} + r_Q Q) \quad (\text{D30})$$

$$\Rightarrow \tilde{L} = L - \frac{r_Q}{r_L}(Q - \tilde{Q}) \quad (\text{D31})$$

Calculating quality-adjusted years of schooling thus requires data on changes in quality ($Q - \tilde{Q}$) and the relative value of quality (r_Q) compared to quantity (r_L). I now turn to estimating each of these two components.

Estimation of Global Trends in Conditional Literacy The first step is to estimate ($Q - \tilde{Q}$), the evolution of quality of schooling for the working-age population from 1980 to 2019. The database of [Le Nestour, Moscoviz, and Sandefur \(2022\)](#) provides information on literacy at grade 5 in 86 countries for two cohorts born during the 1952-1999 period (see [Le Nestour, Moscoviz, and Sandefur, 2022](#), Table 7). Starting from these two data points by country, I estimate average conditional literacy for the working-age population.

First, I divide all figures by 5, so that the indicator corresponds to expected literacy per year of education. Second, I linearly interpolate and extrapolate this indicator backwards and forwards, to cover all cohorts born from 1915 to 1994. This is a very conservative assumption: it amounts to considering that education quality continued to decline at the same pace after the last cohort observed, and was already declining at the same pace from 1915 until the first cohort observed. Third, I construct measures of average education quality of the working-age population by averaging the indicator over all cohorts aged 25 to 65 in a given year, weighted by the population of each cohort. Data on population by age is taken from the United Nations' World Population Prospects. Finally, I impute the indicator for missing countries. To be as conservative as possible, I assume that quality in missing countries has declined at the speed of India, that is, at a very fast

pace. I view this last case as an extreme and implausible scenario, given above-mentioned evidence on the stability or rise of test scores in many countries.

Estimation of Returns to Literacy The second step is to estimate r_Q/r_L , the return to literacy relative to a year of schooling. This requires data on personal income, years of schooling, and literacy at the individual level. I was able to find four high-quality surveys covering these three variables: the Brazilian 2015 PNAD survey, the Indonesian 1998 SUSENAS survey, the Pakistani 2018 HIES survey, and the South African 2019 GHS survey. In each of these four countries, I estimate the relative returns to literacy by running two regressions: a regression relating the log of total personal income to literacy, and a regression relating the log of total personal income to years of schooling, controlling for gender, age, and age squared in each case. I restrict the sample to workers with either no schooling or basic education, to make sure that the two estimates are comparable (nearly all workers with more than basic education are literate).

The results are presented in Table D6. Returns to schooling range from 3% to 6% per year of basic education, while returns to literacy range from 16 to 34 log points. The ratio between the two coefficients is very similar across countries, ranging from 5 in Pakistan to 7 in Indonesia. I take a value of 6 in what follows. This amounts to assuming that moving the entire population from being illiterate to literate is equivalent to increasing average schooling by 6 years.

Results Finally, I construct measures of quality-adjusted years of schooling. I set 1980 as the benchmark year, and adjust estimates of average years of schooling in all other years from 1981 to 2019 so that they reflect the quality observed in 1980. For instance, quality-adjusted years of schooling in 2019 are calculated as $\tilde{L}_{2019} = L_{2019} - \frac{r_Q}{r_L}(Q_{2019} - Q_{1980})$, with L_{2019} unadjusted years of schooling observed in 2019, $\frac{r_Q}{r_L} = 6$, and $Q_{2019} - Q_{1980}$ the change in expected literacy per year of schooling from 1980 to 2019. This approach thus amounts to “deflating” years of schooling observed from 1981 to 2019 to express them in 1980 equivalents.

Figure D4 compares the share of growth explained by global income percentile before and after making this adjustment. The two lines are very close to each other: even under the very conservative assumptions on the decline in education quality outlined above, the main results are almost unchanged. The share of growth explained by education declines by about 2 to 10 percentage points depending on the percentile considered, with the greatest changes observed at the upper-middle of the distribution. The results presented in this paper thus appear to be strongly robust to potential changes in education quality observed since 1980.

Table D1 – IV Estimates of the Returns to Schooling

Source	Country	Policy	Level	Gender	OLS β	IV β	OLS SE	IV SE
Selected Studies								
Leigh and Ryan (2008)	Australia	CSL	Secondary	A	12.8	11.8	.5	3.5
Lemieux and Card (2001)	Canada	EXP	Tertiary	A	7	8	.2	4.4
Oreopoulos (2006b)	Canada	CSL	Secondary	A	9.9	9.6	.7	2.5
Fang et al. (2012)	China	CSL	Secondary	A	9	20	.4	.6
Huang and Zhu (2022)	China	EXP	Tertiary	M	4.9	16.5	.2	2.8
Assaad et al. (2023)	Egypt	CSL	Primary	A	2.1	3.8	.3	4.5
Brunello, Weber, and Weiss (2015)	Europe	CSL	Secondary	A	4.2	5.6	.3	2.6
Khanna (2023)	India	EXP	Primary	A	10	15.5		
Carneiro, Lokshin, and Umapathi (2017)	Indonesia	EXP	Secondary	A	9	12.9	.5	4.8
Duflo (2001)	Indonesia	EXP	Primary	A	7.7	10.6	.06	2.2
Denny and Harmon (2000)	Ireland	EXP	Secondary	A	7.9	13.6	.6	2.5
Kuwait (2024)	Kuwait	EXP	Secondary	M	8.6	5	.03	.2
Fabregas and Navarro-Sola (2024)	Mexico	EXP	Secondary	A	19	19		
Oyelere (2010)	Nigeria	EXP	Primary	A	2.6	2.7	.1	1.3
Aakvik, Salvanes, and Vaage (2010)	Norway	CSL	Secondary	M	8.2	9.4	.1	.2
Bhuller, Mogstad, and Salvanes (2017)	Norway	CSL	Secondary	A	9.3	11.2	.2	4.8
Sakellariou (2006)	Philippines	EXP	Secondary	M	6.1	11.4		
Portugal et al. (forthcoming)	Portugal	CSL	Secondary	A	8.54	9.34	.01	.06
Vieira (1999)	Portugal	CSL	Primary	A	7.8	5.3		
Kyui (2016)	Russian Federation	EXP	Tertiary	A	5.5	16.1	.2	1
Meghir and Palme (1999)	Sweden	CSL	Secondary	A	2.8	3.6	.7	2.1
Spohr (2003)	Taiwan	CSL	Secondary	M	5.4	5.8	.01	2.6
Zhang (2020)	Taiwan	CSL	Secondary	A	6.7	5.6	.01	.1
Delesalle (2021)	Tanzania	EXP	Primary	A	2.6	5.7	.01	2.1
Korwatanasakul (2023)	Thailand	CSL	Primary	A	10.9	7.64	.17	.75

Patrinos, Psacharopoulos, and Tansel (2020)	Turkey	CSL	Primary	M	8.3	18		
Torun (2018)	Turkey	CSL	Primary	M	2.4	1.7	1	1
Clay, Lingwall, and Stephens (2021)	USA	CSL	Secondary	M	8.4	7.7	.1	1.5
Li (2024)	USA	CSL	Secondary	W	11.2	16.9	.1	2.3
Stephens and Yang (2014)	USA	CSL	Secondary	A	6.3	-1.4		
Zimmerman (2014)	USA	THR	Tertiary	A	10	14		
Patrinos and Sakellariou (2005)	Venezuela	CSL	Secondary	A	5.2	8.6		
Vu and Vu-Thanh (2022)	Viet Nam	EXP	Tertiary	A	11.2	19		
Other Studies								
Alzúa, Gasparini, and Haimovich (2015)	Argentina	CSL	Secondary	A		15.8		4.2
Powdthavee, Lekfuangfu, and Wooden (2013)	Australia	CSL	Secondary	M		15.5		3.3
Eble and Hu (2019)	China	CSL	Primary	A		5.2		
Fan et al. (2018)	China	THR	Tertiary	A		6.6		3.2
Brunello, Fort, and Weber (2009)	Europe	CSL	Secondary	M		5.1		
Hsiao (2023)	Indonesia	EXP	Primary	A		26		
Hicks and Duan (2023)	Jordan	CSL	Secondary	A		5.2		
Liwiński (2020)	Poland	CSL	Secondary	A		13.1		4.7
Fischer et al. (2019)	Sweden	CSL	Primary	A		2.4		.6
Meghir and Palme (2005)	Sweden	CSL	Secondary	A		4.8		
Aydemir and Kirdar (2017)	Turkey	CSL	Secondary	M		2.5		
Oreopoulos and Salvanes (2011)	USA	CSL	Secondary	M		13.1		.6
Excluded Studies								
Filmer and Schady (2014)	Cambodia	THR	Secondary	A		20.8		340.8
Domnisoru (2021)	France	CSL	Secondary	M		5.4		1.7
Grenet (2013)	France	CSL	Secondary	M	7.3	-.4	.1	2.9
Pischke (2007)	Germany	CSL	Secondary	A				
Pischke and Wachter (2008)	Germany	CSL	Secondary	A	7.4	1.6	.1	1.5
Duflo, Dupas, and Kremer (2024)	Ghana	EXP	Secondary	A				

Oosterbeek and Webbink (2007)	Netherlands	CSL	Secondary	A				
Buscha and Dickson (2012)	United Kingdom	CSL	Secondary	A	7.1		9.2	
Buscha and Dickson (2015)	United Kingdom	CSL	Secondary	A	7			
Clark (2023)	United Kingdom	CSL	Secondary	A	.01		2.3	
Delaney and Devereux (2019)	United Kingdom	CSL	Secondary	A	6			
Devereux and Hart (2010)	United Kingdom	CSL	Secondary	A	3			
Dickson (2013)	United Kingdom	CSL	Secondary	A	4.6	10.2	.3	5.1
Dolton and Sandi (2017)	United Kingdom	CSL	Secondary	M	6			
Grenet (2013)	United Kingdom	CSL	Secondary	M	9.5	6.9	.1	2.9
Harmon and Walker (1995)	United Kingdom	CSL	Secondary	A	6.1	15.3	.1	1.5
Oreopoulos (2006a)	United Kingdom	CSL	Secondary	A	8.3	14.8	.1	4.6

Notes. The table reports OLS and IV estimates of the returns to schooling from a collection of existing studies. Policy: IV return to schooling derived from the analysis of compulsory schooling laws (CSL), other educational expansion policies (EXP), or college admission thresholds (THR). Gender: return to schooling estimated in a sample of both men and women (A), men only (M), or women only (W). OLS/IV β : OLS/IV return per year of education. OLS/IV SE: OLS/IV coefficient standard error.

Table D2 – Empirical Estimates of the Elasticity of Substitution Between Skill Groups

Source	Country	Tertiary/Below	Secondary/Below
Long-run elasticity			
Bils, Kaymak, and Wu (2022)	Cross-Country	4 to 6	4 to 6
Hendricks and Schoellman (2023)	Cross-Country	4.5	7.8
Short-run elasticity			
Bowlus et al. (2021)	United States	5.3	
Autor, Goldin, and Katz (2020)	United States	1.62	
Hershbein, Kearney, and Pardue (2020)	United States	1.4	
Acemoglu and Autor (2011)	United States	1.6	
Goldin and Katz (2007)	United States	1.6	2 to 5
Ciccone and Peri (2005)	United States	1.5	
Heckman, Lochner, and Taber (1998)	United States	1.4	
Katz and Murphy (1992)	United States	1.41	
Murphy, Riddell, and Romer (1998)	Canada	1.36	
Angrist (1995)	Palestine	2	
Vu and Vu-Thanh (2022)	Vietnam	2.67	
Fernández and Messina (2018)	Latin America	1.25	2.3
Khanna (2023)	India		4.24
Caselli and Coleman (2006)	Cross-Country	1.3	

Notes. The table reports selected estimates of the elasticity of substitution between skill groups from various empirical studies. [Bils, Kaymak, and Wu \(2022\)](#): unique elasticity of substitution for all skill groups. [Khanna \(2023\)](#): primary/below.

Table D3 – Distributional Growth Accounting, World, 1980-2019:
With Physical Capital Affected by Education

	Annual Income Growth (%)	Contribution of Education (pp.)	Share of Growth Explained (%)
Full Population	1.6%	1.0	62%
Bottom 50%	2.4%	1.3	51%
Bottom 20%	1.9%	1.3	67%
Next 30%	2.6%	1.2	49%
Middle 40%	1.5%	1.3	84%
Top 10%	1.6%	0.8	50%
Top 1%	2.0%	0.7	32%
Top 0.1%	2.5%	0.7	26%

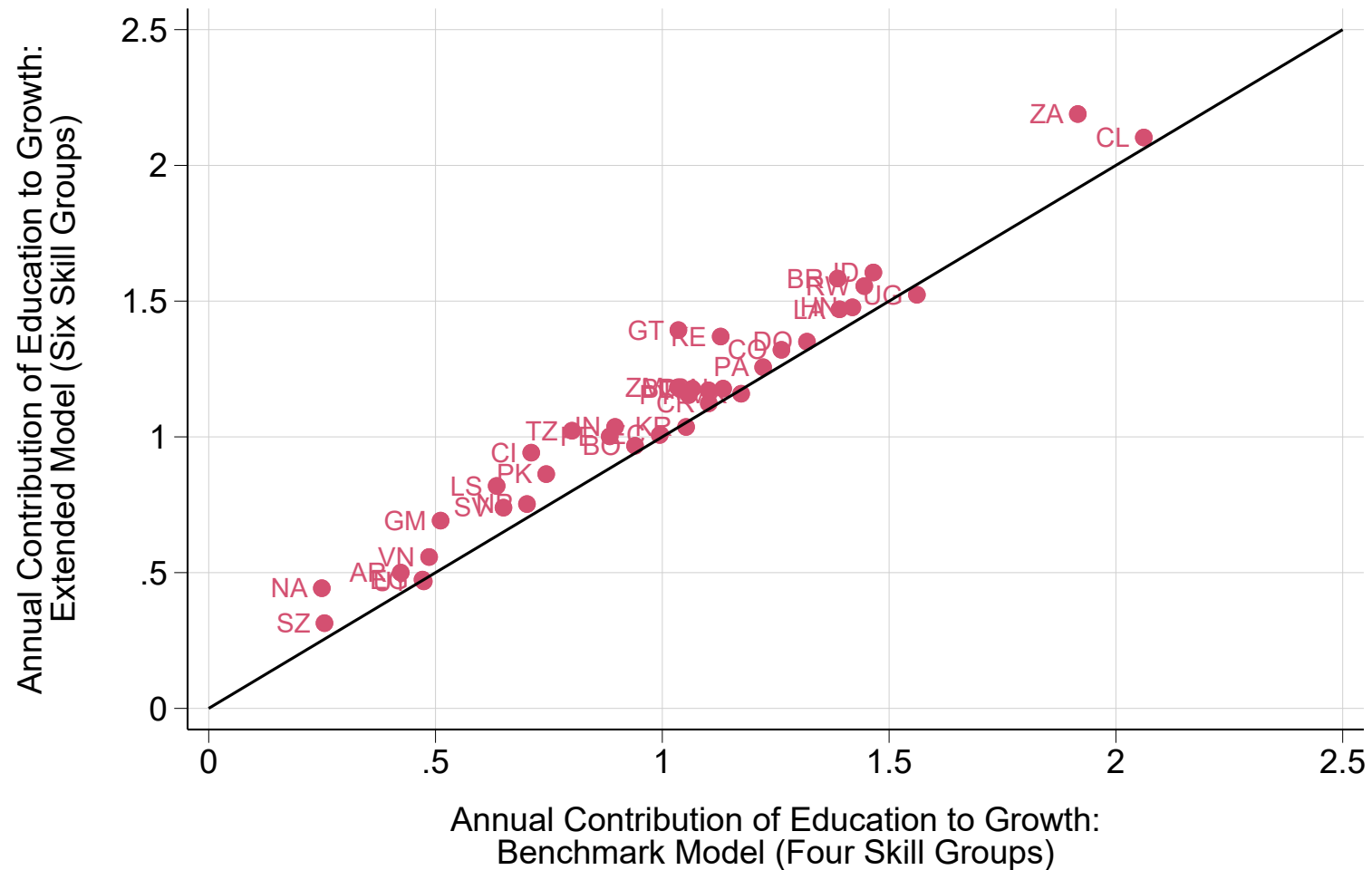
Notes. The table reports actual real annual income growth rates, the contribution of education to growth, and the corresponding share of growth explained by education for different groups of the world distribution of income. Physical capital is assumed to adjust in the counterfactual so as to keep the capital-to-output ratio constant in each country.

Table D4 – Complete Growth Decomposition by World Region, 1980-2019

	Actual Growth	Education	Physical Capital	Residual
Average Country	3.1%	0.7%	1.2%	1.1%
Europe / Northern America	1.6%	0.8%	0.4%	0.4%
Latin America	0.6%	0.8%	0.5%	-0.7%
China	6.1%	0.8%	2.7%	2.6%
India	4.0%	0.6%	1.5%	1.9%
Other Asia-Pacific	3.0%	0.7%	1.1%	1.2%
Middle East and North Africa	1.8%	0.5%	1.5%	-0.2%
Sub-Saharan Africa	1.0%	0.8%	0.1%	0.1%

Notes. The table decomposes average annual income growth into the total contribution of education, the contribution of physical capital, and residual growth by world region. Population-weighted averages of growth rates and contributions of education and physical capital in each country.

Figure D1 – Accounting for Within-Category Variations in Human Capital
Benchmark versus Extended Model with Six Skill Groups: Contribution of Education to Growth



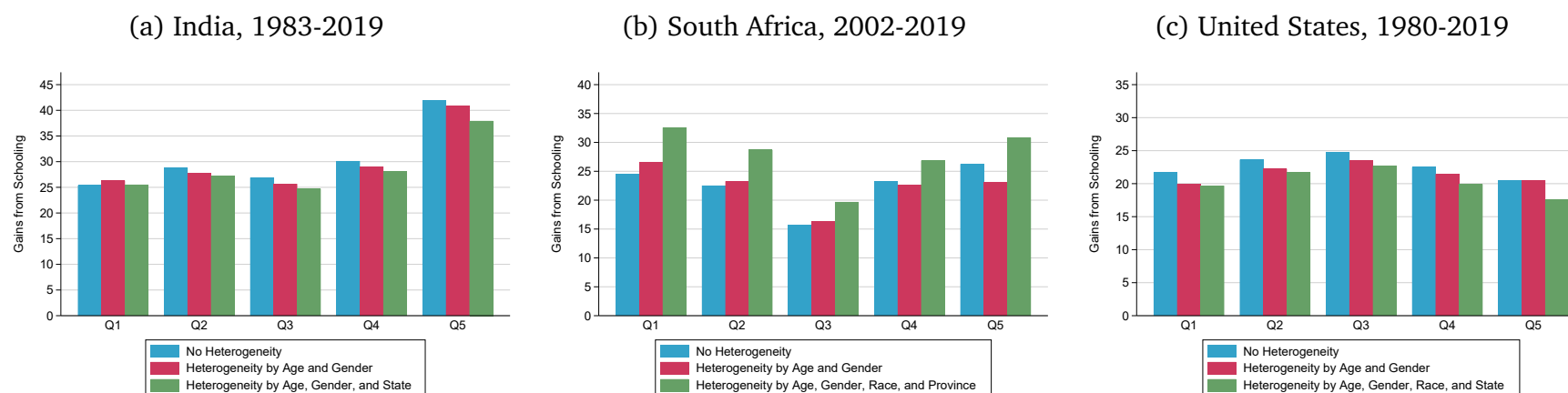
Notes. The table compares the annual contribution of education to economic growth estimated when using the benchmark model versus an extended model with six skill groups, distinguishing workers with incomplete versus complete primary and secondary education. The contribution of education to annual real income growth is equal to the annualized log difference between actual and counterfactual human capital in each country: $\frac{1}{39} \times (\ln H(A^{2019}, L^{2019}) - \ln H(A^{2019}, L^{1980}))$.

Table D5 – Accounting for Within-Category Variations in Human Capital
Benchmark versus Extended Model with Six Skill Groups: Results by Country

	Returns to Schooling		Contribution of Education to Annual Growth (pp.)		Share of Growth Explained by Education (%)	
	Complete Primary	Complete Secondary	Benchmark Model	Extended Model	Benchmark Model	Extended Model
Average Country	14%	27%	1.0	1.1	39%	43%
Argentina	19%	26%	0.4	0.5	61%	72%
Bangladesh	9%	36%	1.1	1.2	22%	24%
Bolivia	31%	3%	0.9	1.0	46%	47%
Brazil	35%	22%	1.4	1.6	>100%	>100%
Chile	8%	19%	2.1	2.1	49%	50%
China	21%	32%	1.1	1.2	14%	14%
Colombia	20%	34%	1.3	1.3	53%	55%
Costa Rica	20%	28%	1.1	1.1	36%	37%
Cote d'Ivoire	17%	71%	0.7	0.9	58%	77%
Dominican Republic	19%	8%	1.3	1.4	27%	27%
Ecuador	30%	17%	1.0	1.0	94%	95%
Egypt	2%	0%	0.5	0.5	8%	8%
El Salvador	31%	17%	0.6	0.7	49%	55%
Eswatini	19%	85%	0.3	0.3	12%	14%
Gambia	14%	23%	0.5	0.7	>100%	>100%
Guatemala	32%	59%	1.0	1.4	>100%	>100%
Honduras	52%	48%	1.4	1.5	>100%	>100%
India	3%	19%	0.9	1.0	14%	17%
Indonesia	17%	35%	1.5	1.6	28%	31%
Kenya	37%	45%	1.1	1.4	>100%	>100%
Lao	7%	11%	1.4	1.5	18%	19%
Lesotho	13%	51%	0.6	0.8	33%	42%
Mexico	30%	28%	1.2	1.2	>100%	>100%
Namibia	25%	92%	0.2	0.4	28%	50%
Nepal	4%	10%	0.7	0.8	19%	20%
Pakistan	3%	32%	0.7	0.9	26%	30%
Panama	29%	37%	1.2	1.3	23%	24%
Paraguay	19%	23%	1.1	1.2	44%	48%
Peru	9%	15%	0.9	1.0	39%	45%
Rwanda	22%	31%	1.4	1.6	49%	53%
South Africa	13%	42%	1.9	2.2	>100%	>100%
South Korea	10%	18%	1.1	1.0	15%	15%
Tanzania	24%	73%	0.8	1.0	32%	41%
Thailand	19%	16%	1.1	1.2	19%	20%
Uganda	23%	104%	1.6	1.5	45%	44%
Uruguay	29%	29%	0.5	0.5	15%	15%
Vietnam	9%	29%	0.5	0.6	7%	8%
Zambia	21%	55%	1.0	1.2	>100%	>100%

Notes. The table compares growth accounting results obtained when using the benchmark model versus an extended model with six skill groups, distinguishing workers with incomplete versus complete primary and secondary education. Returns to schooling: estimated returns to complete relative to incomplete primary education, and to complete relative to incomplete secondary education. Contribution of education to annual growth: $\frac{1}{39} \times (\ln H(A^{2019}, L^{2019}) - \ln H(A^{2019}, L^{1980}))$.

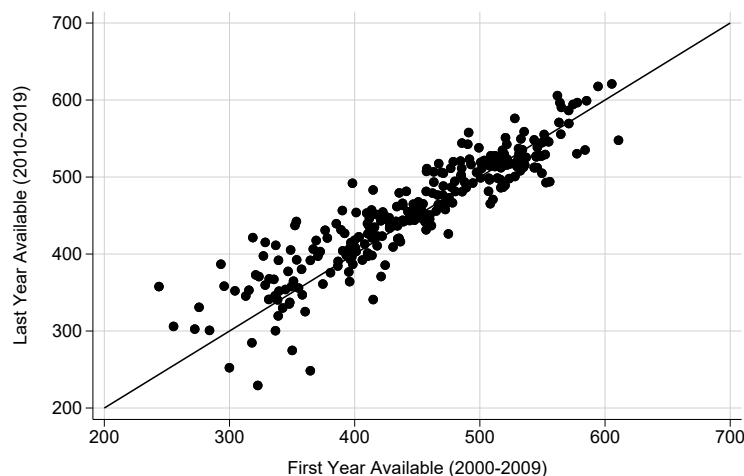
Figure D2 – Accounting for Heterogeneous Educational Expansion:
Gains from Schooling With and Without Heterogeneous Educational
Expansion by Socioeconomic Characteristic



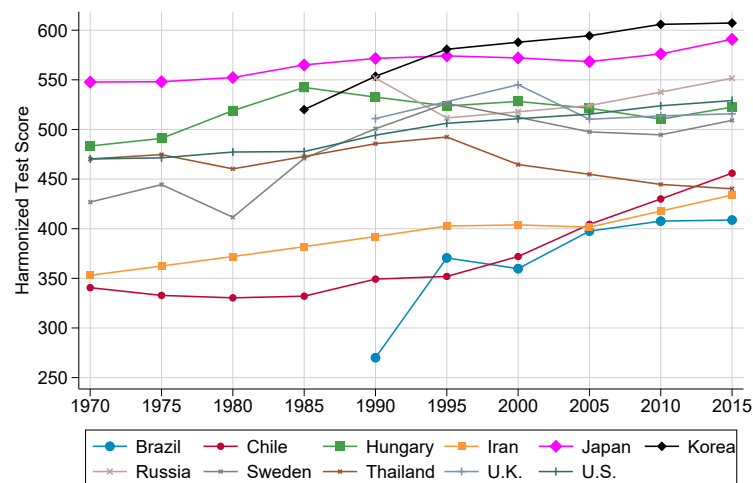
Notes. The figure compares gains from education by income quintile before and after accounting for heterogeneous educational expansion by socioeconomic characteristic in India, South Africa, and the United States. In each case, education levels of individuals are downgraded without any heterogeneity (specification 1), by age-gender cell (specification 2), or by age-gender-region-race or age-gender-region cell (specification 3) until reaching counterfactual levels. Their earnings are then reduced using estimates of returns to schooling by level. Finally, the figure plots schooling gains by income quintile, defined as the percent difference between actual income and counterfactual income absent educational expansion. India: educational attainment by age, gender, and state of residence in 1983 estimated using the 1983 National Sample Survey; simulation run on the 2019 Periodic Labor Force Survey. South Africa: educational attainment by age, gender, race, and province estimated using the 2002 General Household Survey; simulation run on the 2019 General Household Survey. United States: educational attainment by age, gender, state, and race estimated using the 1980 Current Population Survey; simulation run on the 2019 Current Population Survey.

Figure D3 – Accounting for Education Quality: Empirical Evidence on the Evolution of Education Quality

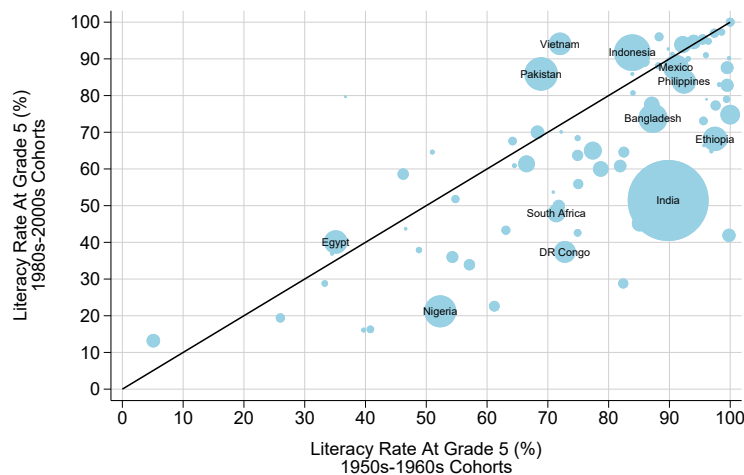
(a) Harmonized Test Scores: 2000-2009 vs. 2010-2019



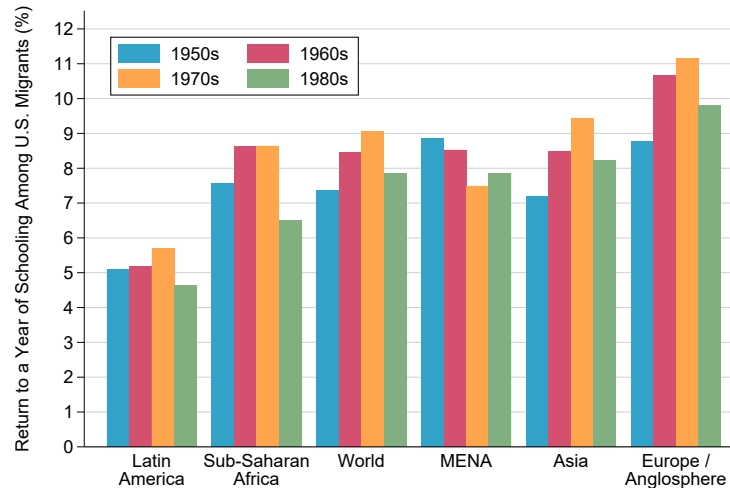
(b) Long-Run Trends in Test Scores in Selected Countries



(c) Literacy at Grade 5: 1950-1960 versus 1980-2000 Cohorts



(d) Trends in Returns to Schooling Across Cohorts of U.S. Migrants



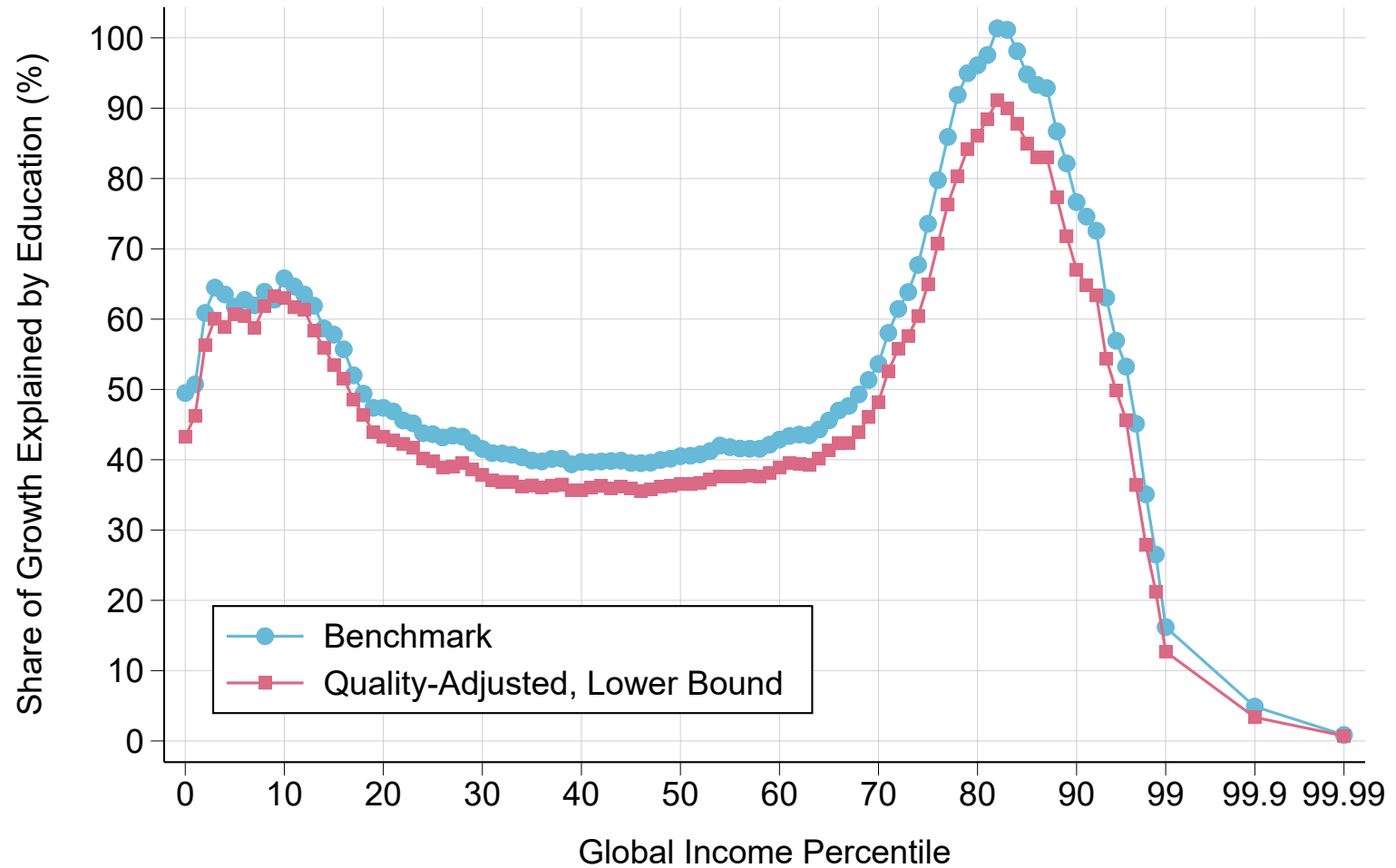
Notes. The figure provides empirical evidence on the evolution of education quality from various data sources. Panel (a) compares average test scores in 2000-2009 and 2010-2019 based on data from [Angrist et al. \(2021\)](#). Each point corresponds to a test score reported for a given country $\times$ education level (primary/secondary) $\times$ subject (maths/science/reading). Panel (b) plots the evolution of harmonized test scores in selected countries from 1970 to 2015, based on data from [Altinok, Angrist, and Patrinos \(2018\)](#). Panel (c) compares expected literacy at age 5 among 1950s-1960s cohorts (x-axis) and 1980s-2000s cohorts (y-axis), drawing on estimates from [Le Nestour, Moscoviz, and Sandefur \(2022\)](#). Panel (d) plots the return to a year of schooling among U.S. migrants coming from each world region, based on data from U.S. censuses and American Community Surveys. Population-weighted average of returns to schooling estimated for a given country of origin $\times$ decade of birth cell.

Table D6 – Accounting for Education Quality:
Returns to Literacy

	Brazil	Indonesia	Pakistan	South Africa
Return to Literacy	0.34*** (0.01)	0.34*** (0.01)	0.26*** (0.01)	0.16*** (0.05)
Return to Schooling	0.06*** (0.00)	0.05*** (0.00)	0.05*** (0.00)	0.03*** (0.01)
Literacy / Schooling	5.75	7.38	5.31	6.48

Notes. The table reports estimates of returns to literacy, returns to schooling, and the ratio between the two. The coefficient on literacy corresponds to a regression of the log of personal income on literacy; the coefficient on years of schooling corresponds to a separate regression of the log of personal income on years of schooling. Both regressions control for gender, age, and age squared in each country. Data sources: 2015 Brazil PNAD survey, 1998 Indonesia SUSENAS survey, 2018 Pakistan HIES survey, 2019 South Africa GHS survey. Standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure D4 – Accounting for Education Quality:
Share of Growth Explained by Education: Benchmark Versus Lower Bound on Decline in Education Quality



Notes. The figure plots the share of growth explained by education by global income percentile, before and after adjusting educational attainment for potential changes in education quality. Quality-adjusted estimates correct years of schooling for the decline in education quality estimated by [Le Nestour, Moscoviz, and Sandefur \(2022\)](#), so that years of schooling are expressed in 1980 equivalents throughout the period. Countries with missing data are attributed the decline in quality observed in India.

E. Data Appendix: Survey Microdata

E.1. Main Survey Microdatabase

The survey microdata used in the main analysis come from five main data sources. Table [E1](#) reports the names and years of surveys used in each country.

ILO Microdata The main data source is a set of harmonized household surveys that were collected and compiled by the International Labor Organization. These surveys were run by statistical institutes and are nationally representative. The majority of them are labor force surveys explicitly designed to measure labor market variables. Other surveys are typically household income and expenditure surveys, or multi-purpose surveys that collected information on a range of indicators together with labor market variables. I use the survey closest to 2019 in each country. The database presents itself as a single harmonized microfile. The main variables are country, year, household ID, sample weight, wage income (from main job, second job, and all jobs combined), self-employment income (from main job, second job, and all jobs combined), age, gender, education, labor force participation, occupation (ISCO-08), industry, and rural-urban location. I define personal income as the sum of all wage and self-employment income received by an individual. The sample is restricted to all individuals with strictly positive personal income.

European Statistics on Income and Living Conditions Although the ILO data do cover European countries, the coding of educational attainment is broader than in the original microfiles, so I decide to rely on my own data collection. The European Statistics on Income and Living Conditions (EU-SILC) cover detailed information on personal income and education in 32 countries from 2003 to 2020. I harmonize EU-SILC surveys in the same way as those of the ILO, defining personal income as the sum of individual wage and mixed income. I then replace all ILO surveys by this microfile, with the exception of France, Portugal, and Switzerland, for which the ILO provides national labor force surveys of even better quality.

I2D2 Database The I2D2 database consists in a set of harmonized surveys that have been compiled by the World Bank since the 1990s. Many surveys are common to the ILO and I2D2 database, but they differ in two dimensions. First, the ILO prioritizes labor force surveys, while the I2D2 database is more focused on household income and expenditure surveys. Second, the ILO microdata has better coverage of recent years (the I2D2 was discontinued in 2017), while the I2D2 database has better historical coverage. In the

main analysis, I use the I2D2 to cover 11 additional countries that are not in the ILO database: Azerbaijan, Gabon, Georgia, Haiti, Iran, Kyrgyzstan, Kosovo, Kazakhstan, North Macedonia, Saudi Arabia, and Ukraine.

Life in Transition Survey For Belarus, Montenegro, Uzbekistan, I rely on the Life in Transition Survey (LITS). The LITS is far from being ideal, with sample sizes of only 3,000-5,000, yet it is to the best of my knowledge the only data source available to measure individual incomes and education in these three countries. I use the last wave of the LITS, fielded in 2016, which I harmonize in the same way as the ILO.

Country-Specific Surveys Finally, I collect and harmonize surveys from country-specific data portals to cover 11 additional countries: China, Iraq, India, Japan, Mozambique, Morocco, Russia, Somalia, South Africa, South Korea, South Sudan, Tunisia, and the United States.

For seven countries, I was available to find and harmonize a high-quality survey providing detailed information on individual incomes. This type of survey was available for China (2018 Chinese Household Income Project), India (2019 Periodic Labor Force Survey), Russia (2019 Russia Longitudinal Monitoring Survey), South Korea (2019 Korean Labor and Income Panel Study), Tunisia (2014 Labor Force Survey), South Africa (2019 General Household Survey), and the United States (2019 Current Population Survey).

For six other countries, I rely on surveys of lower quality or only providing information on household expenditure. For Japan, in the absence of better publicly available data, I use the 2017 general household survey, which does cover individual income and education but has a small sample size (about 1,000). I use household income and expenditure surveys for Iraq (Household Socio-Economic Survey), Mozambique (Inquérito aos orçamentos familiares), Morocco (Household Expenditure Survey), Somalia (High Frequency Survey), and South Sudan (High Frequency Survey), which provide information on individual employment and education, as well as total household expenditure, but not on individual incomes. In the absence of better information, I proxy personal income by splitting equally household expenditure among adults in employment, excluding unemployed or inactive individuals as well as children.

Data Quality The surveys used in each country vary in their focus and sample size, which makes them a more or less ideal source to measure the relationship between education and individual incomes. In Table E1, I classify the quality of each survey as high, medium, or low. High-quality surveys are surveys with detailed labor and individual income modules and large sample sizes (102 countries representing 61% of the world's population). Medium-quality surveys are multi-purpose surveys that collected information on

both labor market variables and other dimensions of households' economic conditions (43 countries, 32% of the world's population). Low-quality surveys are surveys with low sample sizes and/or no information on personal income (9 countries, 4% of the world's population).

E.2. Historical Survey Microdatabase

In section V, I estimate skill-biased technical change since 2000 in 109 countries. The surveys used in this part of the analysis come from four main sources.

The primary source is the I2D2 database, which allows me to cover the distribution of education and income in 63 countries around 2000.

The second source is the Global Labor Database maintained by the World Bank, which is similar to I2D2 but specifically focused on labor force surveys. It allows me to cover 11 additional countries: Brazil, Chile, Colombia, Indonesia, India, Mongolia, Nepal, the Philippines, Pakistan, Thailand, and Turkey. I also rely on the Global Monitoring Database, the successor of the I2D2, to cover Bhutan, and on the ILO microdatabase to cover Canada and Cambodia.

The third source is EU-SILC, which allows me to cover 29 European countries. While EU-SILC was first fielded in 2003, personal income and educational attainment suffer from measurement problems in the earlier waves of the survey, so I use the 2006 wave. Unfortunately, this is the only survey available when it comes to accurately measuring incomes and education in the European Union in the 2000s.

Finally, I rely on country-specific surveys for China (2002 Chinese Household Income Project), Russia (2000 Russia Longitudinal Monitoring Survey), and the United States (2000 Current Population Survey).

The complete list of surveys is reported in Table [E2](#), together with a quality indicator derived in the same way as in Table [E1](#).

Table E1 – Survey Data Sources: Main Analysis

Country	Source	Survey Year	Quality
Europe			
Albania	Living Standards Survey	2012	Medium
Austria	EU Statistics on Income and Living Conditions	2019	High
Belarus	Life in Transition Survey	2016	Low
Belgium	EU Statistics on Income and Living Conditions	2019	High
Bosnia and Herzegovina	Labour Force Survey	2016	High
Bulgaria	EU Statistics on Income and Living Conditions	2019	High
Croatia	EU Statistics on Income and Living Conditions	2019	High
Czechia	EU Statistics on Income and Living Conditions	2019	High
Denmark	EU Statistics on Income and Living Conditions	2019	High
Estonia	EU Statistics on Income and Living Conditions	2019	High
Finland	EU Statistics on Income and Living Conditions	2019	High
France	Employment Survey	2019	High
Germany	EU Statistics on Income and Living Conditions	2019	High
Greece	EU Statistics on Income and Living Conditions	2019	High
Hungary	EU Statistics on Income and Living Conditions	2019	High
Iceland	EU Statistics on Income and Living Conditions	2018	High
Ireland	EU Statistics on Income and Living Conditions	2019	High
Italy	Labour Force Survey	2019	High
Latvia	EU Statistics on Income and Living Conditions	2019	High
Lithuania	EU Statistics on Income and Living Conditions	2019	High
Luxembourg	EU Statistics on Income and Living Conditions	2019	High
Malta	EU Statistics on Income and Living Conditions	2019	High
Moldova	Labour Force Survey	2019	High
Montenegro	Life in Transition Survey	2016	Low
Netherlands	EU Statistics on Income and Living Conditions	2019	High
North Macedonia	Labor Force Survey	2019	High
Norway	EU Statistics on Income and Living Conditions	2019	High
Poland	EU Statistics on Income and Living Conditions	2019	High
Portugal	Employment Survey	2019	High
Romania	EU Statistics on Income and Living Conditions	2019	High
Russia	Russia Longitudinal Monitoring Survey	2019	Medium
Serbia	Labour Force Survey	2019	High
Slovakia	EU Statistics on Income and Living Conditions	2019	High
Slovenia	EU Statistics on Income and Living Conditions	2019	High
Spain	EU Statistics on Income and Living Conditions	2019	High
Sweden	EU Statistics on Income and Living Conditions	2019	High
Switzerland	Labour Force Survey	2019	High
Ukraine	Household Living Conditions Survey	2019	Medium
United Kingdom	Labour Force Survey	2018	High
Northern America			
Canada	Labour Force Survey	2019	High
USA	Current Population Survey	2019	High
Latin America			
Argentina	Permanent Household Survey, Urban	2019	Medium
Barbados	Survey on Living Conditions	2016	Medium
Belize	Labour Force Survey	2019	High
Bolivia	Continuous Employment Survey	2019	High
Brazil	Continuous National Household Sample Survey	2019	High
Chile	National Survey on Socio-Economic Conditions	2017	High
Colombia	Integrated Household Survey	2019	High
Costa Rica	National Household Survey	2019	Medium
Dominican Republic	Continuous National Labour Force Survey	2019	High
Ecuador	National Survey on Employment	2019	High
El Salvador	Multi-purpose Household Survey	2019	Medium
Guatemala	Monthly Employment and Income Survey	2019	High
Guyana	Labour Force Survey	2019	High
Haiti	Enquête sur les Conditions de Vie des Ménages	2012	Medium

Honduras	Continous Multi-Purpose Household Survey	2019	Medium
Jamaica	Labour Force Survey	2014	High
Mexico	National Occupation and Employment Survey	2019	High
Nicaragua	National Household Survey on Measuring Living Conditions	2014	Medium
Panama	Labour Market Survey	2019	High
Paraguay	Continous Household Survey	2017	High
Peru	National Household Survey	2019	Medium
Suriname	Survey on Living Conditions	2016	Medium
Trinidad and Tobago	Continuous Sample Survey of the Population	2016	Medium
Uruguay	Continous Household Survey	2019	Medium
Venezuela	Household Sample Survey	2017	Medium
Asia			
Afghanistan	Households Living Conditions Survey	2014	Medium
Australia	Household, Income and Labour Dynamics Survey	2019	High
Bangladesh	Labour Force Survey	2017	High
Bhutan	Labour Force Survey	2019	High
Brunei Darussalam	Labour Force Survey	2014	High
Cambodia	Labour Force Survey	2019	High
China	China Household Income Project	2018	Medium
Fiji	Employment, Unemployment Survey	2016	High
India	Periodic Labour Force Survey	2019	High
Indonesia	National Labour Force Survey	2019	High
Iran	Household Expenditure and Income Survey	2019	Medium
Japan	General Social Survey	2017	Low
Kazakhstan	Household Budget Survey	2017	Medium
Kosovo	Labor Force Survey	2017	High
Kyrgyzstan	Kyrgyz Integrated Household Survey	2017	Medium
Lao	Labour Force Survey	2017	High
Maldives	Household Income and Expenditure Survey	2019	Medium
Mongolia	Labour Force Survey	2019	High
Myanmar	Labour Force Survey	2019	High
Nepal	Labour Force Survey	2017	High
Pakistan	Labour Force Survey	2019	High
Philippines	Labour Force Survey	2018	High
South Korea	Korean Labor and Income Panel Study	2019	High
Sri Lanka	Labour Force Survey	2018	High
Tajikistan	Living Standards Survey	2009	Medium
Thailand	Household Socio-Economic Survey	2019	Medium
Timor-Leste	Labour Force Survey	2016	High
Tonga	Labour Force Survey	2018	High
Uzbekistan	Life in Transition Survey	2016	Low
Vietnam	Labour Force Survey	2019	High
Middle East and North Africa			
Armenia	Household Labour Force Survey	2019	High
Azerbaijan	Monitoring Survey for Social Welfare	2015	Medium
Cyprus	EU Statistics on Income and Living Conditions	2019	High
Egypt	Labour Force Sample Survey	2018	High
Georgia	Household Integrated Survey	2019	Medium
Iraq	Household Socio-Economic Survey	2012	Low
Jordan	Employment and Unemployment Survey	2019	High
Lebanon	Labour Force Survey	2019	High
Morocco	Household Expenditure Survey	2014	Low
Palestine	Labour Force Survey	2019	High
Saudi Arabia	Household Expenditure and Income Survey	2013	Medium
Sudan	Household Survey	2011	Medium
Tunisia	Labor Force Survey	2014	High
Turkey	Household Labour Force Survey	2019	High
Yemen	Labour Force Survey	2014	High
Sub-Saharan Africa			
Angola	Employment Survey	2019	High
Benin	Integrated Survey of Household Living Conditions	2018	Medium
Botswana	Multi-Topic Household Survey	2019	Medium

Burkina Faso	Regional Integrated Survey on Employment and the Informal Sector	2018	High
Burundi	Living Standards Survey	2014	Medium
Cabo Verde	Continuous Multi-Objective Survey	2015	Medium
Cameroon	Household Survey	2014	Medium
Chad	Modular and Integrated Household Survey on Living Conditions	2018	Medium
Comoros	National Survey on Employment and the Informal Sector	2014	High
Côte d'Ivoire	National Survey on the Employment Situation	2019	High
DR Congo	Survey on Employment and household's living conditions	2012	High
Djibouti	Djiboutian Household Survey	2017	Medium
Eswatini	Labour Force Survey	2016	High
Ethiopia	National Labor Force Survey	2013	High
Gabon	Survey on Poverty Evaluation and Monitoring	2017	Medium
Gambia	Labour Force Survey	2018	High
Ghana	Labour Force Survey	2015	High
Guinea	National Survey on Employment and the Informal Sector	2019	High
Guinea-Bissau	Harmonized Survey on Household Living Conditions	2018	Medium
Kenya	Household Budget Survey	2019	Medium
Lesotho	Labour Force Survey	2019	High
Liberia	Labour Force Survey	2017	High
Madagascar	National Survey on Employment and the Informal Sector	2015	High
Malawi	Labour Force Survey	2013	High
Mali	Continuous Household Employment Survey	2018	High
Mauritania	Living Standards Survey	2019	Medium
Mauritius	Continuous Multi-Purpose Household Survey	2019	Medium
Mozambique	Inquérito aos orçamentos familiares	2014	Low
Namibia	Labour Force Survey	2018	High
Niger	National Survey on Household Living Conditions	2014	Medium
Nigeria	Socio Economic Survey	2019	Medium
Republic of the Congo	Employment Survey	2009	High
Rwanda	Labour Force Survey	2017	High
Senegal	National Employment Survey	2019	High
Sierra Leone	Integrated Household Survey	2018	Medium
Somalia	High Frequency Survey	2017	Low
South Africa	General Household Survey	2019	High
South Sudan	High Frequency Survey	2015	Low
Tanzania	National Household Budget Survey	2012	Medium
Togo	Regional Integrated Survey on Employment and the Informal Sector	2017	High
Uganda	National Labour Force Survey	2017	High
Zambia	Labour Force Survey	2019	High
Zimbabwe	Labour Force Survey	2014	High

Table E2 – Survey Data Sources, 1990s-2000s

Country	Source	Survey Year	Quality
Europe			
Austria	EU Statistics on Income and Living Conditions	2006	High
Belarus	Household Living Standards Survey	2000	Medium
Belgium	EU Statistics on Income and Living Conditions	2006	High
Bosnia and Herzegovina	Living Standards Measurement Study	2001	Medium
Bulgaria	EU Statistics on Income and Living Conditions	2006	High
Czechia	EU Statistics on Income and Living Conditions	2006	High
Denmark	EU Statistics on Income and Living Conditions	2006	High
Estonia	EU Statistics on Income and Living Conditions	2006	High
Finland	EU Statistics on Income and Living Conditions	2006	High
France	EU Statistics on Income and Living Conditions	2006	High
Germany	EU Statistics on Income and Living Conditions	2006	High
Greece	EU Statistics on Income and Living Conditions	2006	High
Hungary	EU Statistics on Income and Living Conditions	2006	High
Iceland	EU Statistics on Income and Living Conditions	2006	High
Ireland	EU Statistics on Income and Living Conditions	2006	High
Italy	EU Statistics on Income and Living Conditions	2006	High
Latvia	EU Statistics on Income and Living Conditions	2006	High
Luxembourg	EU Statistics on Income and Living Conditions	2006	High
Malta	EU Statistics on Income and Living Conditions	2006	High
Moldova	Household Budget Survey	2001	Medium
Netherlands	EU Statistics on Income and Living Conditions	2006	High
North Macedonia	Household Budget Survey	2000	Medium
Norway	EU Statistics on Income and Living Conditions	2006	High
Poland	EU Statistics on Income and Living Conditions	2006	High
Portugal	EU Statistics on Income and Living Conditions	2006	High
Romania	EU Statistics on Income and Living Conditions	2006	High
Russia	Russia Longitudinal Monitoring Survey	2000	Medium
Serbia	Household Budget Survey	2004	Medium
Slovakia	EU Statistics on Income and Living Conditions	2006	High
Slovenia	EU Statistics on Income and Living Conditions	2006	High
Spain	EU Statistics on Income and Living Conditions	2006	High
Sweden	EU Statistics on Income and Living Conditions	2006	High
Switzerland	EU Statistics on Income and Living Conditions	2006	High
Ukraine	Household Living Conditions Survey	2002	Medium
United Kingdom	EU Statistics on Income and Living Conditions	2006	High
Northern America			
Canada	Labor Force Survey	2000	High
USA	Current Population Survey	2000	High
Latin America			
Argentina	Permanent Household Survey	2000	Medium
Barbados	Labor Force Survey	1996	High
Belize	Labor Force Survey	1999	High
Bolivia	Continuous Household Survey	2000	Medium
Brazil	National Household Sample Survey	1999	High
Chile	Chile National Socioeconomic Characterization Survey	2000	High
Colombia	Integrated Household Survey	2000	High
Costa Rica	Multipurpose Household Survey	2000	Medium
Dominican Republic	Labor Force Survey	2000	High
Ecuador	Labor Force Survey	2000	High
El Salvador	Multipurpose Household Survey	2000	Medium
Guatemala	Household Living Conditions Survey	2000	Medium
Guyana	Household Survey of Living Conditions	1999	Medium
Honduras	Multi-Purpose Permanent Household Survey	1999	Medium
Jamaica	Survey of Living Conditions	1999	Medium
Mexico	National Survey of Household Income and Expenditure	2000	Medium
Panama	Household Survey	2000	Medium
Paraguay	Permanent Household Survey	1999	High

Peru	National Household Survey	2000	Medium
Suriname	Expenditure Household Survey	1999	Medium
Trinidad and Tobago	IPUMS	2000	Medium
Uruguay	Continuous Household Survey	2000	Medium
Venezuela	Household Sampling Survey	2000	Medium
Asia			
Australia	Household, Income and Labour Dynamics Survey	2002	High
Bangladesh	Labor Force Survey	1999	High
Bhutan	Bhutan Living Standard Survey	2003	Medium
Cambodia	Labor Force Survey	2000	High
China	China Household Income Project	2002	Medium
India	Employment and Unemployment Survey	1999	High
Indonesia	National Labour Force Survey	2000	High
Iran	Household Budget Survey	2004	Medium
Kazakhstan	Household Budget Survey	2001	Medium
Lao	Lao Expenditure and Consumption Survey	2002	Medium
Maldives	Vulnerability and Poverty Assessment Survey	1998	Medium
Mongolia	Labor Force Survey	2002	High
Nepal	Labor Force Survey	1998	High
Pakistan	Labor Force Survey	1999	High
Philippines	Labor Force Survey	2001	High
South Korea	Labor Force Survey	2001	High
Thailand	Labor Force Survey	2000	High
Timor-Leste	Population and Health Survey	2001	Medium
Vietnam	Household Living Standards Survey	2002	Medium
Middle East and North Africa			
Armenia	Integrated Living Conditions Survey	1998	Medium
Cyprus	EU Statistics on Income and Living Conditions	2006	High
Egypt	Egypt Labour Market Panel Survey	1998	High
Georgia	Household Integrated Survey	2000	Medium
Jordan	Household Income and Expenditure Survey	2002	Medium
Lebanon	Living Standards Survey	2004	Medium
Morocco	Household Living Standards Survey	1998	Medium
Palestine	Labor Force Survey	2000	High
Turkey	Labor Force Survey	2002	High
Yemen	Household Budget Survey	1998	Medium
Sub-Saharan Africa			
Angola	Household Income and Expenditure Survey	2000	Medium
Botswana	Household Income and Expenditure Survey	2002	Medium
Burkina Faso	Enquête Prioritaire	1998	Medium
Burundi	Enquête Prioritaire	1998	Medium
Chad	Consumption and Informal Sector Survey	2003	Medium
Côte d'Ivoire	Household Living Standards Survey	2002	Medium
Djibouti	Djibouti Household Survey	1996	Medium
Eswatini	Household Income and Expenditure Survey	2000	Medium
Gambia	High Frequency Phone Survey	1998	Medium
Ghana	Living Standards Survey	1998	Medium
Madagascar	Enquête Harmonisée sur les Conditions de Vie des Ménages	1999	Medium
Mauritania	Permanent Survey of Living Conditions of Households	2000	Medium
Mauritius	Continuous Multi-Purpose Household Survey	1999	Medium
Nigeria	Living Standards Survey	2003	Medium
Rwanda	Integrated Household Living Conditions Survey	2000	Medium
Senegal	Household Survey	2001	Medium
Sierra Leone	Integrated Household Survey	2003	Medium
South Africa	Labor Force Survey	2000	High
Uganda	National Household Survey	1999	Medium
Zambia	Living Conditions Monitoring Survey	1998	Medium

F Data Appendix: Returns to Schooling

F.1. OLS Estimates of the Returns to Schooling

In the main analysis, I use estimates of returns to schooling by level estimated in each country. I rely on the following modified Mincerian equation:

$$\ln y_i = \alpha + \beta_{pri}D_{i,pri} + \beta_{sec}D_{i,sec} + \beta_{ter}D_{i,ter} + X_i\beta + \varepsilon_i \quad (F32)$$

With y_i earned income of individual i in a given country, $D_{i,pri}$, $D_{i,sec}$, and $D_{i,ter}$ dummies for having reached primary, secondary, and tertiary education, and X_i a vector of controls including gender, an age quartic, and interactions between gender and the age quartic. Earned income is the sum of all wage and self-employment income received by a given individual. I restrict the sample to all individuals aged 25 or above with strictly positive income. I estimate this regression separately in each country and extract estimates of β_{pri} , β_{sec} , and β_{ter} . In some surveys, there are too few observations to estimate the return to a specific education level. I choose to set returns as missing when there are less than 100 observations covering each education level—for instance, the return to primary education is set as missing if there are either less than 100 workers with no schooling or less than 100 workers with primary education. This is the case of 66 countries for the return to primary education and 24 countries for the return to secondary education. I then fill in missing values by assuming that the ratio of the missing return to that of the return to the category above is the same as in the average country. This imputation makes little difference to my results, since missing returns are those for which the share of workers with the corresponding education levels is particularly low.

One may be concerned that the estimated returns might be sensitive to alternative empirical specifications. Table F1 investigates the robustness of my results to using a standard Mincerian equation with only gender, age, and age squared as controls, as well as to restricting the dependent variable to wage income. The estimated returns are almost identical across these different specifications.

Another concern is that workers and self-employed individuals declaring positive personal income might only represent a subset of the population. This is particularly concerning in low-income countries, where a large fraction of the population often relies on subsistence agriculture and thus ends up excluded from my estimation of the returns to schooling. I investigate this concern in Appendix Table F2 by comparing

three specifications. The first one corresponds to a standard Mincerian equation estimated at the individual level, restricting the sample to individuals declaring positive personal income. The second specification corresponds to a “household-level Mincerian equation,” regressing per-capita expenditure on adults’ average years of schooling. The third specification repeats the second specification, but after restricting the sample to households with at least one adult declaring positive personal income, which is useful to check whether the results are driven by selection into reporting positive income. I estimate these returns for eleven countries characterized by high poverty rates and large agricultural sectors: India, Pakistan, the Democratic Republic of the Congo, Burkina Faso, Côte d’Ivoire, Guinea-Bissau, Mali, Niger, Sénégal, and Togo. For each of these eleven countries, I was able to collect and manually harmonize survey microdata covering personal income, household expenditure, and educational attainment.

The three estimates fall very close to each other, with a Mincerian return typically varying from 7% to 10%. Individual returns are slightly higher than household-level returns in some countries. This is to be expected given that variations in consumption are more driven by other factors, such as savings and transfers received by other households and the government. Yet, there are also countries where individual returns are lower, such as Burkina Faso and Mali. Household-level returns before and after excluding households with no reported income are almost identical. Together, these findings provide reassuring evidence that the returns estimated in this paper provide a good approximation of the true returns to schooling for the population as a whole.

F.2. OLS Estimates of Returns Per Year of Schooling

F.2.1. Estimation Using the I2D2 Database

Figure II suggests that the returns to schooling are convex. The ILO microdata unfortunately do not allow studying heterogeneity in returns to schooling per year, because it does not cover information on completed years of schooling within each education category. For instance, the return to primary education could be low mainly because most of workers with some primary education have one or two years of schooling.

Returns Per Year of Schooling by Level To shed light on this issue, I turn to the I2D2 database. The surveys gathered in I2D2 are often the same as those used in the rest of this paper. Surveys fielded around

2019 and covering exact years of schooling are available for 62 countries. I estimate the following regression:

$$\ln y_{ic} = \alpha + \zeta^{pri} S_{ic}^{pri} + \zeta^{sec} S_{ic}^{sec} + \zeta^{ter} S_{ic}^{ter} + X_{ic} \beta + \mu_c + \varepsilon_{ic} \quad (F33)$$

Where S_{ic}^{pri} , S_{ic}^{sec} , and S_{ic}^{ter} are completed years of primary (0-6 years), secondary (7-12 years), and tertiary (above 12 years) education, respectively. X_{ic} is a vector of controls including gender, an age quartic, and interactions between gender and the age quartic. μ_c are country fixed effects. ζ^{pri} , ζ^{sec} , and ζ^{ter} can thus be interpreted as yearly returns to completing the first six years, next six years, and above twelve years of education. I run this regression separately by world region and for all countries together. I find that returns to schooling are strongly convex: the return per year of schooling is greater at higher levels of education in all world regions (see Table II).

Table F3 present results of a similar Mincerian equation but with only completed years of schooling. Table F4 also reports results of a similar regression, except that the explanatory variables are years of schooling and years of schooling squared. The coefficient on years of schooling squared is positive and statistically significant at the 1% level in all world regions.

Returns to Completed Degrees Finally, Table F5 reports results of the following regression:

$$\ln y_i = \alpha + \kappa_{pri1} D_{i,pri1} + \kappa_{pri2} D_{i,pri2} + \kappa_{sec1} D_{i,sec1} + \kappa_{sec2} D_{i,sec2} + \kappa_{ter1} D_{i,ter1} + \kappa_{ter2} D_{i,ter2} + X_i \beta + \mu_c + \varepsilon_i \quad (F34)$$

Where $D_{i,pri1}$, $D_{i,pri2}$, $D_{i,sec1}$, $D_{i,sec2}$, $D_{i,ter1}$, and $D_{i,ter2}$ are dummies for having reached incomplete primary, complete primary, incomplete secondary, complete secondary, incomplete tertiary, and complete tertiary education. This specification allows deriving total returns to completing a whole education cycle as:

$$\kappa_{pri} = D_{i,pri2} \quad (F35)$$

$$\kappa_{sec} = D_{i,sec2} - D_{i,pri2} \quad (F36)$$

$$\kappa_{ter} = D_{i,ter2} - D_{i,sec2} \quad (F37)$$

In other words, κ_{pri} is the return to completing primary education versus having no schooling, κ_{sec} is the return to completing secondary education versus having completed primary education, and κ_{ter} is the

return to completing tertiary education versus having completed secondary education. For the definitions of complete vs. incomplete degrees, I rely on the coding of these variables that is directly reported in I2D2.

This regression is less straightforward to interpret, given that (1) primary, secondary, and tertiary cycles differ in length and (2) these lengths can vary significantly across countries. In particular, primary and secondary education typically take 6 years, while most individuals in low- and middle-income countries do not go beyond three years of tertiary education.

Table F5 reports the results of the regression. The corresponding estimates of κ_{pri} , κ_{sec} , and κ_{ter} are displayed at the bottom of the table. In all regions, the total return to primary education is lower than the total return to tertiary education, despite the latter cycle being much shorter in the majority of countries.

F2.2. Comparison With the Previous Literature

The convexity of returns to schooling contrasts with much of the macroeconomics literature in development and growth accounting, which often assumes decreasing returns to human capital. To understand this discrepancy, it is useful to go back to previous studies documenting or assuming decreasing returns. Among seminal studies, [Hall and Jones \(1999\)](#) assume a return to the first four years of education of 13.4%, a return of 10.1% for the next four years, and a return of 6.8% above eight years. These figures correspond to average returns per year of schooling in Sub-Saharan Africa, the world as a whole, and OECD countries, respectively, which they take from a previous review of the literature. They do not use actual data on returns to schooling by level. [Bils and Klenow \(2000\)](#) adopt the same approach, concluding from the negative cross-country correlation between Mincerian returns and average years of schooling that there are diminishing returns to human capital. Unfortunately, cross-country differences in average returns may not be a good proxy for heterogeneity in returns by level within each country. In fact, returns to schooling do vary substantially across regions and are on average higher in low-income countries, despite being convex in every region.

Another influential study is [Montenegro and Patrinos \(2021\)](#), who exploit the I2D2 database to estimate returns to schooling in a large sample of countries from 1970 to 2014. Their analysis suggests that returns per year of schooling are highest for tertiary education and lowest for secondary education, with primary education falling in between. However, their calculation of annual returns by level contains a mistake. They estimate total returns by level using a specification similar to equation 11, but then divide each coefficient by the number of years of schooling in each category instead of taking the exponent. For instance, they

estimate the return to a year of primary education as $\gamma_c^{pri} = \beta_c^{pri} / L^{pri}$, with L^{pri} the number of years of primary education, while the true formula should be $\gamma_c^{pri} = \exp\left(\beta_c^{pri}\right)^{1/L^{pri}} - 1$. This might explain why they find higher returns to primary education than to secondary education, in contrast with my analysis using the same database. Another reason might be that they restrict the analysis to wage earners, while my estimates cover both wage earners and the self-employed.

My results are more consistent with recent estimates of the returns to schooling. For instance, [Jedwab et al. \(2023\)](#) document much higher returns per year of schooling for workers with more than 13 years of schooling in the I2D2 database. Similarly, [Rossi \(2022\)](#) finds clear evidence of convex returns to schooling in a sample of twelve developed and developing countries.

Table F1 – Returns to Schooling: Sensitivity to Alternative Specifications

	(1)	(2)	(3)	(4)
Primary	0.157*** (0.007)	0.153*** (0.007)	0.135*** (0.010)	0.135*** (0.010)
Secondary	0.524*** (0.006)	0.511*** (0.006)	0.543*** (0.007)	0.532*** (0.007)
Tertiary	1.079*** (0.007)	1.054*** (0.007)	1.069*** (0.008)	1.047*** (0.008)
Extended Model	No	Yes	No	Yes
Depvar = Log Wage	No	No	Yes	Yes
N	5,103,826	5,103,826	3,839,134	3,839,134
Adj. R-squared	0.82	0.82	0.88	0.88

Notes. The table reports estimates of total returns to schooling, comparing models with “standard” versus “extended” controls, as well as models using the log of personal income versus the log of wages as the dependent variable. Standard model: controls for gender, age, and age squared. Extended model: controls for gender, an age quartic, and interactions between gender and the age quartic, as in [Lemieux \(2006\)](#). Depvar = log wage: the dependent variable is the log of wages rather than the log of total personal income. Pooled regression across all countries. All estimates include country fixed effects. Observations are weighted to match each country’s total population. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F2 – Returns to Schooling: Personal Income Versus Per-Capita Consumption

	Individual Income	Consumption All Households	Consumption Households With Income Only
India	0.070*** (0.001)	0.063*** (0.001)	0.063*** (0.001)
Pakistan	0.074*** (0.001)	0.071*** (0.001)	0.071*** (0.001)
DR Congo	0.075*** (0.002)	0.071*** (0.002)	0.070*** (0.002)
Burkina Faso	0.089*** (0.006)	0.106*** (0.003)	0.086*** (0.005)
Benin	0.078*** (0.003)	0.068*** (0.002)	0.071*** (0.003)
Côte d'Ivoire	0.073*** (0.003)	0.058*** (0.002)	0.052*** (0.002)
Guinea-Bissau	0.041*** (0.004)	0.061*** (0.002)	0.052*** (0.003)
Mali	0.072*** (0.005)	0.080*** (0.002)	0.061*** (0.003)
Niger	0.108*** (0.004)	0.103*** (0.003)	0.101*** (0.003)
Sénégal	0.078*** (0.003)	0.074*** (0.002)	0.075*** (0.002)
Togo	0.100*** (0.006)	0.078*** (0.002)	0.076*** (0.005)

Notes. The table compares returns to schooling estimated with three specifications. The first specification regresses individual income on individual years of schooling, controlling for age, gender, and their interaction. The second specification regresses per-capita consumption on average years of schooling of working-age adults at the household level, controlling for household size, average age, and the share of women. The third specification does the same, but after restricting the sample to households with at least one adult declaring positive personal income. India: 2019 PLFS survey. Pakistan: 2018 HIES survey. DR Congo: 2012 ECM survey. Other countries: 2018 EHCVM surveys. Robust standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F3 – Returns Per Year of Schooling: Pooled Regression Results

	(1) All Countries	(2) Latin America	(3) China	(4) India	(5) Other Asia	(6) MENA	(7) Sub-Saharan Africa
Years of Schooling	0.076*** (0.000)	0.096*** (0.000)	0.092*** (0.002)	0.063*** (0.001)	0.057*** (0.001)	0.020*** (0.000)	0.097*** (0.001)
N	1,285,895	781,577	30,036	122,870	119,548	78,881	152,983
R Squared	0.78	0.88	0.21	0.35	0.91	0.74	0.79

Notes. The table reports results of pooled regressions relating the log of personal income to years of schooling. The full sample covers 62 countries with information on exact completed years of schooling. All estimates include country fixed effects and control for gender, an age quartic, and interactions between gender and the age quartic. Observations are weighted to match the total population of each country.

Table F4 – Returns Per Year of Schooling: Pooled Regression Results with Years of Schooling Squared

	(1) All Countries	(2) Latin America	(3) China	(4) India	(5) Other Asia	(6) MENA	(7) Sub-Saharan Africa
Years of Schooling	0.002* (0.001)	0.075*** (0.001)	0.052*** (0.008)	-0.020*** (0.002)	0.011*** (0.001)	-0.001 (0.001)	0.005 (0.004)
Years of Schooling Squared	0.004*** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.005*** (0.000)	0.003*** (0.000)	0.001*** (0.000)	0.005*** (0.000)
N	1,285,895	781,577	30,036	122,870	119,548	78,881	152,983
R Squared	0.79	0.88	0.21	0.38	0.91	0.74	0.79

Notes. The table reports results of pooled regressions relating the log of personal income to years of schooling and years of schooling squared. The full sample covers 62 countries with information on exact completed years of schooling. All estimates include country fixed effects and control for gender, an age quartic, and interactions between gender and the age quartic. Observations are weighted to match the total population of each country.

Table F5 – Returns to Detailed Educational Attainment Categories: Pooled Regression Results

	(1) All Countries	(2) Latin America	(3) China	(4) India	(5) Other Asia	(6) MENA	(7) Sub-Saharan Africa
Incomplete Primary	0.063*** (0.014)	0.312*** (0.018)	0.198* (0.105)	0.098*** (0.011)	0.011 (0.016)	0.039 (0.034)	0.056*** (0.022)
Complete Primary	0.196*** (0.014)	0.505*** (0.018)	0.389*** (0.105)	0.144*** (0.014)	0.238*** (0.021)	0.137*** (0.041)	0.132*** (0.022)
Incomplete Secondary	0.391*** (0.014)	0.713*** (0.018)	0.579*** (0.103)	0.233*** (0.010)	0.247*** (0.014)	0.213*** (0.036)	0.393*** (0.021)
Complete Secondary	0.630*** (0.014)	0.921*** (0.017)	0.853*** (0.104)	0.406*** (0.012)	0.499*** (0.015)	0.229*** (0.025)	0.627*** (0.024)
Incomplete Tertiary	0.903*** (0.015)	1.205*** (0.018)	1.042*** (0.105)	0.809*** (0.023)	0.658*** (0.019)	0.289*** (0.038)	0.989*** (0.025)
Complete Tertiary	1.185*** (0.014)	1.648*** (0.017)	1.258*** (0.104)	1.087*** (0.014)	0.811*** (0.016)	0.529*** (0.025)	1.271*** (0.022)
Compl. Primary vs. No Schooling	0.20	0.50	0.39	0.14	0.24	0.14	0.13
Compl. Secondary vs. Compl. Primary	0.43	0.42	0.46	0.26	0.26	0.09	0.49
Compl. Tertiary vs. Compl. Secondary	0.56	0.73	0.40	0.68	0.31	0.30	0.64
N	1,291,371	781,614	30,036	122,870	119,659	78,910	158,282
R Squared	0.89	0.92	0.21	0.37	0.94	0.91	0.82

Notes. The table reports results of pooled regressions relating the log of personal income to dummies taking 1 if the individual has the corresponding level of educational attainment and 0 otherwise. Each coefficient thus captures the return to reaching a specific educational attainment category with respect to having no schooling. Returns to completing full cycles of education are reported at the bottom of the table. The full sample covers 62 countries with detailed educational attainment information. All estimates include country fixed effects and control for gender, an age quartic, and interactions between gender and the age quartic. Observations are weighted to match the total population of each country.

G. Data Appendix: Educational Attainment and Income Distribution Data

G.1. Educational Attainment Data

Barro-Lee Database The primary data source used to measure the evolution of educational attainment is the database compiled by Barro and Lee (2013) and updates.⁶ The database covers the distribution of educational attainment by age group and gender in 146 countries at five year intervals from 1950 to 2015. It covers 123 countries out of the 150 countries studied in this paper. The education categories are no schooling, incomplete primary, complete primary, incomplete secondary, complete secondary, incomplete tertiary, and complete tertiary. I interpolate linearly the share of individuals belonging to each category between missing years, and extrapolate linearly educational attainment by age and gender after 2015, so as to cover the entire 1980-2019 period.

IPUMS and Survey Data For the 27 countries absent from the Barro-Lee database, I rely on census and survey data. For Burkina Faso (1985-2006), Ethiopia (1984-2007), Guinea (1983-2014), and Palestine (1997-2017), the data source is the census microdata samples available from IPUMS International. For India, which is covered by the Barro-Lee database but displays somewhat erratic trends, I rely instead on the education modules of the national sample survey (1983-2017), which I collected and harmonized for the purpose of this paper. For the remaining 22 countries, in the absence of better data, I use cohort-level trends in educational attainment observed in the surveys collected in this paper.⁷ I first aggregate the distribution of educational attainment by cohort and gender in each survey. I linearly interpolate and extrapolate when needed, so as to cover all cohorts born since 1915-1920 (aged 60-65 in 1980). I then derive estimates of educational attainment of the 1980 to 2019 working-age populations by taking the weighted average across cohorts belonging to the working-age population in the corresponding year.

Matching Survey and Aggregate Data To derive accurate estimates of counterfactual income absent educational expansion, it is important to make sure that educational attainment in the survey data matches perfectly aggregate data used to derive the counterfactual. Although education levels do correlate strongly in the two sources, some inconsistencies remain. For instance, aggregate and survey data sometimes report

⁶See <http://www.barrolee.com/>.

⁷The countries are Angola, Azerbaijan, Bosnia and Herzegovina, Bhutan, Belarus, Cabo Verde, Cambodia, Chad, Djibouti, Georgia, Guinea-Bissau, Kosovo, Lebanon, Montenegro, Madagascar, Macedonia, Nigeria, Somalia, Suriname, South Sudan, Timor-Leste, and Uzbekistan.

incomplete degrees as complete and sometimes do not, or code lower secondary education as primary education. To make sure that the two sources coincide, I first manually recode some categories in survey and/or aggregate data, country by country, by visually inspecting the distribution of educational attainment in the two sources. The result of this manual recoding process is displayed in figures [G1a](#), [G1b](#), [G1c](#), and [G1d](#), which compare the share of the working-age population with no schooling, primary/basic education, secondary education, and tertiary education in survey versus aggregate data. The two sources end up very close to each other after recoding.

Second, I perform a final small adjustment to the sample weights of each survey to make sure that education levels by age and gender match perfectly in the two sources. I combine aggregate data on the distribution of attainment with data on total population by age and gender from the UN to derive estimates of the total number of individuals belonging to each of 40 education-age-gender cells. I then use linear calibration to ensure that total weights match the total population belonging to each cell in each country. The result is a new weight variable that ensures that the distribution of educational attainment by age and gender (and for the working-age population as a whole) in the survey data matches perfectly that observed in aggregate data.

G.2. Income Distribution Data

The final step of the distributional growth accounting exercise consists in moving from labor income to total income, and comparing counterfactual to actual real income growth rates. This requires data on the distribution of income in each country, aggregate labor and capital income shares, and the share of income received from labor and capital by income group within each country.

Pretax Income Inequality Data Data on the world distribution of income come from the World Inequality Database (WID). It reports average per-capita pretax income by percentile in all countries in the world from 1980 to 2019. The income concept is pretax national income, that is, total income received by individuals before accounting for taxes and transfers, but after accounting for the operation of pension and unemployment systems. Importantly, all components of net national income (GDP, minus consumption of fixed capital, plus net foreign income) are allocated to individuals, following the Distributional National Accounts (DINA) framework (see [Chancel et al., 2022](#); [Piketty, Saez, and Zucman, 2018](#)). This ensures that all income distributions are consistent with macroeconomic growth rates and aggregate capital and labor income shares recorded in the national accounts. The database is constructed by compiling estimates from

detailed national or regional studies, which combine surveys, tax data, and national accounts to construct distributions that are conceptually comparable across countries (see for instance [Piketty, Saez, and Zucman \(2018\)](#) on the United States, [Blanchet, Chancel, and Gethin \(2022\)](#) on Europe, and [De Rosa, Flores, and Morgan \(2022\)](#) on Latin America).

Aggregate Labor and Capital Income Shares Aggregate factor income shares come from [Bachas et al. \(2022\)](#), who combine a number of sources to build a new database on the components of net national income worldwide since 1965. Their database provides a decomposition of net domestic product into compensation of employees, mixed income, the operating surplus of households (actual and imputed rental income), and the operating surplus of corporations (profits net of depreciation).

I define the labor income share as the share of income attributable to compensation of employees and mixed income. This is the definition of the labor share that is the most conceptually meaningful in my context, given that my microdata cover individual income and returns to schooling for both wage earners and the self-employed. In the main analysis, I thus make the conservative assumption that human capital only affects wages and mixed income, while leaving capital income unchanged.

Capital Income Concentration The last step is to estimate how the capital income share varies alongside the income distribution in each country. Data on this decomposition are scarce. Drawing on selected high-quality studies, I was able to derive such profiles for the United States ([Piketty, Saez, and Zucman, 2018](#)), South Africa ([Chatterjee, Czajka, and Gethin, 2022](#)), and 10 Latin American countries ([De Rosa, Flores, and Morgan, 2022](#)). The corresponding series are plotted in Appendix Figure G2. The profiles look very similar across these three cases. The capital share is always below 20% for the bottom 90% of earners, corresponding mostly to imputed rental income. It rises exponentially at the very top of the distribution, where the main source of income is from bonds and stock. Given these similarities, I use the average profile observed across countries, which I rescale in each country-year to match the aggregate capital income share.

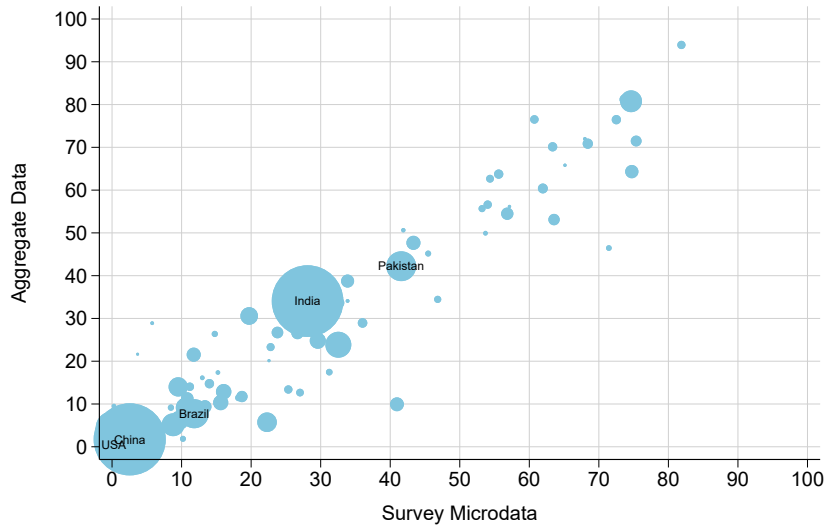
World Bank Income Distribution Data I also report results using World Bank consumption and income distribution data instead of the World Inequality Database. The database is available from the World Bank's Poverty and Inequality Platform (PIP), and presents itself in the form of distributions available for selected countries and years. The income concept is either consumption or posttax disposable income per capita, depending on the country. All values are reported in 2017 PPP USD per day.

Unfortunately, although the World Bank regularly reports indicators of global poverty, it does not publish underlying estimates of the world distribution of income. I thus attempt to reconstruct measures of pretax and posttax income myself. Starting from available data, I first extrapolate the average income of each country-percentile to missing years using real GDP per capita growth rates. For the 17 countries entirely missing, I use estimates from the World Inequality Database. The resulting database yields trends in global poverty almost identical to those officially reported by the World Bank. I then reconstruct measures of pretax income. In the absence of any information on savings, I define pretax income in each country as consumption or disposable income, minus social assistance transfers, plus direct taxes. The distribution of social assistance transfers and direct taxes come from two companion papers ([Fisher-Post and Gethin, 2023](#); [Gethin, 2024](#)).

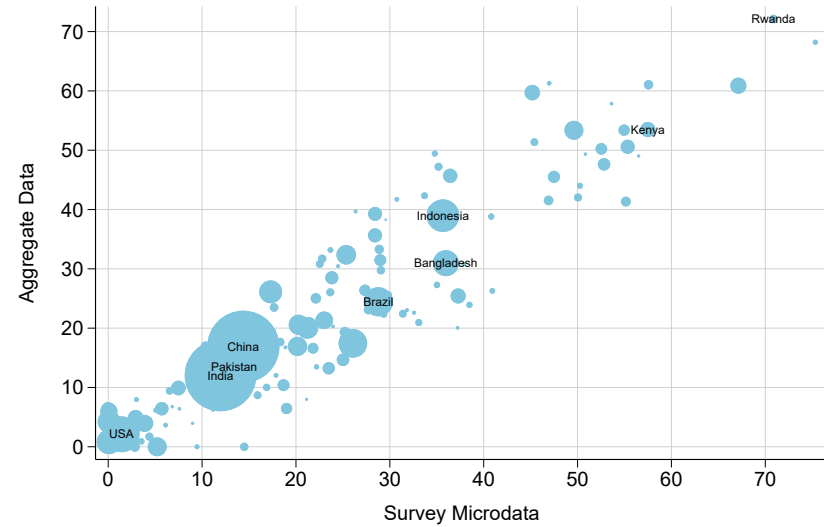
Both the levels and trends in global poverty in the WID data differ from those of the World Bank for at least four main reasons. First, World Bank estimates mostly focus on consumption (posttax disposable income minus net household saving), while my focus here is on income. The difference between consumption and income can be large, with major implications for levels and trends in inequality in some cases (see for instance [Chancel et al., 2023](#)). Second, the estimates presented here are consistent with national income growth rates, while World Bank estimates are based on surveys and do not attempt to bridge gaps between survey and national accounts aggregates. Third, some of the estimates used in this paper are based on detailed country-specific studies that rely on data sources that may differ from those of the World Bank in a number of countries, including China ([Piketty, Yang, and Zucman, 2019](#)), India ([Chancel and Piketty, 2019](#)), and Latin America ([De Rosa, Flores, and Morgan, 2022](#)). See [Chancel and Piketty \(2021\)](#). Fourth, I use GDP purchasing power parity conversion factors, while the World Bank only corrects for price differences in household final consumption expenditure. For all these reasons, I view the World Inequality Database as a more adequate source for conducting the particular analysis developed in this paper. A more systematic comparison of these two datasets is left to future research.

Figure G1 – Distribution of Educational Attainment in Barro-Lee Versus Survey Data (% of Working-Age Population)

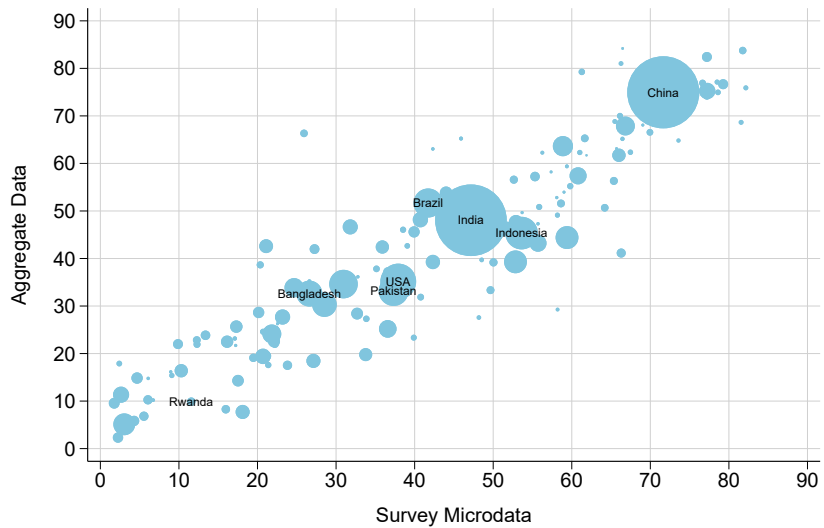
(a) No Schooling



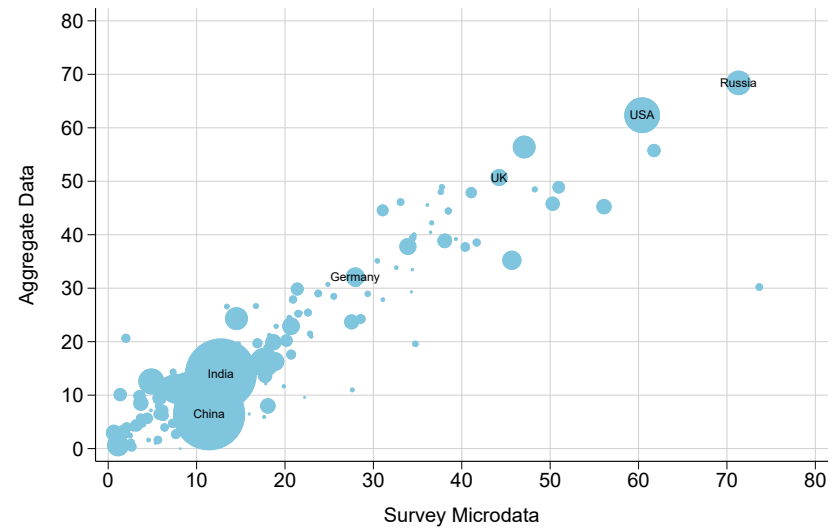
(b) Primary



(c) Secondary

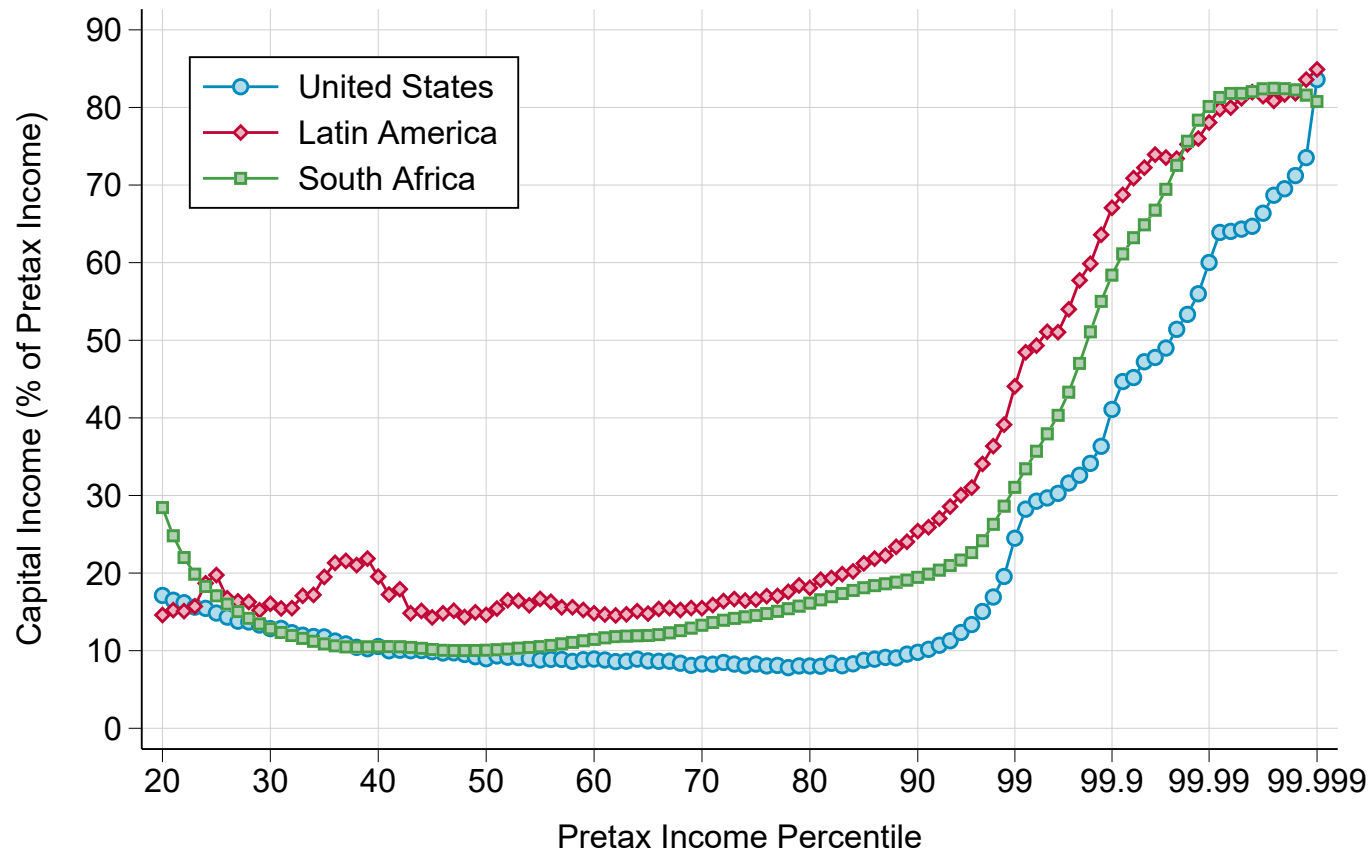


(d) Tertiary



Notes. The figure compares estimates of the share of the working-age population with no schooling, primary, secondary, and tertiary education in the survey microdata (x-axis) and aggregate data from Barro and Lee (2013) and other sources (y-axis), after manual reclassification of educational categories in each country.

Figure G2 – The Concentration of Capital Income
in the United States, Latin America, and South Africa



Notes. The figure plots the capital income share by pretax income percentile in the United States, South Africa, and Latin America in 2019. Capital income is defined as all income other than compensation of employees and mixed income. Author's elaboration combining data from [Piketty, Saez, and Zucman \(2018\)](#) for the United States, [De Rosa, Flores, and Morgan \(2022\)](#) for Latin America, and [Chatterjee, Czajka, and Gethin \(2021\)](#) for South Africa.

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